INDIGENE

STUDY CONSOLE

GST 103

Use of Library & ICT

The Complete A-Grade Study Guide

- Every chapter of GST 103, rebuilt in plain language
- Alzheimer’s Mode openers · Exam hacks · Trap alerts
- 20 trap questions after each chapter, with reasons
- FUOYE website, portal & email section
- Full marked past-question bank at the end

Read it once. Do the traps. Then run the CBT.

How to use this guide

★ ALZHEIMER'S MODE (the simple version)

Do not read this like a novel. Read one chapter, finish its 20 trap questions, then move to the next. When you finish the whole guide, run the GST 103 CBT on the INDIGENE Study Console.

- **Gold boxes** are Alzheimer's Mode: the simple story that makes a topic stick.
- **Orange boxes** are exam hacks: the shortcuts that save time in the hall.
- **Red boxes** are trap alerts: where examiners trick you.
- **Grey boxes** are definitions: learn these word for word if the question starts with "define".
- Past questions are marked at the end exactly as in the question bank.

How the GST 103 exam usually plays

- It is **4-option multiple choice** and the options are mixed up, so understand the idea; do not memorise a position (A, B, C or D).
- Many options look alike. The **exact word** usually decides the winner (e.g. recto, verso, gazette, gazetteer).

- Numbers and lists get tested: **6 types of library, 4 components, 4 acquisition methods, 13-digit ISBN, 49 pages, 3 x 5 card**.
- Do not choose an answer just because it is the longest. Read each option fully.
- Spend about one minute per question. Skip hard ones and come back.

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CHAPTER 1

Use of Library (Basics, Types, Stock & Rules)

1.1 What is a library?

★ ALZHEIMER'S MODE (the simple version)

Think of a library like your own bedroom wardrobe, but for knowledge. Clothes are not just thrown in a heap; they are arranged so you can find a shirt in seconds. A library is a **place where information is collected, arranged, kept safe and given out to people who need it**. No arrangement = not a library, just a pile of books.

The word **library** comes from the Latin word **"liber"**, which means **book**. Some people think a library is just a collection of books in a room with no organisation, others think it is only a reading room. Both ideas are too small.

KEY DEFINITION

Library (Librarians' Glossary) — a collection of books and other library materials kept for reading, study and consultation; a place, building, room or rooms set apart for keeping and using a collection of books; and a collection of films, photographs and other visual non-book materials, tapes, discs, computers and programmes.

KEY DEFINITION

Library (Ranganathan's definition) — a public institution or establishment charged with the care of a collection of books, the duty of making them accessible to those who require them, and the task of converting every person in its neighbourhood into a **habitual library goer and reader of books**.

- The library is an **educational resource centre**, an **information centre** and a **store house of knowledge**.
- It is also called a "**nest**" where scholars are hatched.
- Its users include students, lecturers, researchers and other scholars allowed to use it.
- Major roles: **acquisition, organisation (processing), storage and dissemination (giving out)** of information resources.

Why do we study Use of Library (GST 103)?

- So you can use library resources **judiciously** (wisely) and successfully.
- No scholar can succeed without the library, so you must not run away from it.
- GST 103 is a **compulsory general course**: register for it and pass it before you can graduate.
- It is the **foundation (take-off point)** for a successful academic, scholar or researcher.

1.2 The six types of library

There are **six** types. Each is unique, though some functions are shared.

Type	Who it serves / what it is	Examples
Academic	Attached to tertiary institutions (university, polytechnic, college of education). Supports **learning, teaching, research and service**. Users: students, lecturers, researchers, non-teaching staff, community professionals.	FUOYE Library, EKSU Library, Federal Polytechnic Ado-Ekiti Library, College of Education Ikere-Ekiti Library

Type	Who it serves / what it is	Examples
National	The **apex** library of a nation, owned by a sovereign nation. Has **legal deposit right** (right to acquire everything published in the country). Issues **ISBN** and **ISSN** to authors and publishers.	National Library of Nigeria, National Library of Kenya, Library of Congress (USA)
Private	Owned by an individual or an organisation like a church. Stock matches the owner's profession.	A lawyer's own law library; a doctor's private medical library
Public	Serves everybody of all ages without discrimination. Called the "**people's university**". Paid for with tax-payers' money. Usually has a children's section.	State and community libraries
Research / Special	Built for a defined or unique group of specialist users.	IITA Ibadan, CBN Abuja, NIIA Lagos, CRIN Ibadan, NISER Ibadan
School	Attached to nursery, primary and secondary schools. Builds reading habit and skill in pupils.	A primary or secondary school library

> ⚡ **EXAM HACK**
> Six types = **A-N-P-P-R-S**: **A**cademic, **N**ational, **P**rivate, **P**ublic, **R**esearch/Special, **S**chool. Special and Research are the SAME type here, so never count seven. ISBN/ISSN = National library. “People’s university” = Public library.

1.3 Attributes of a library (what makes a place a real library)

Many rooms in Nigerian schools and offices are called “libraries” just because a few books are kept there. That is only a collection of books. A real library must have these **five attributes**:

- A convenient **space** set aside for **no other purpose** but library operations.
- A **dynamic collection** of books and other materials.
- Materials **systematically organised**, stored and preserved for posterity.
- Run by a person with **formal training in librarianship** (Librarian, Library Officer or Teacher-Librarian).
- People allowed to use it at **specified times** (opening and closing hours).

> ⚠ **TRAP ALERT**
> Keeping a collection out of reach of users “negates the concept of library”. A locked room full of books is NOT a library.

1.4 The four components of a library

Component	What it means
Books, serials and non-book materials	The collection (the stock).
Building / room(s)	A designated, quiet space with good lighting, ventilation, fans or air conditioners.
Personnel (staff)	They run the library and turn the collection into service.
Clientele (users)	Users, readers and patrons are the same thing. They are the reason the library exists.

Books, serials and non-book materials

- **Book**: printed sheets fastened together, with **not less than 49 pages** apart from the cover pages, for reading, consultation or reference. Less than 49 pages = **pamphlet**.
- **Serials**: publications issued in successive parts, with no planned end. Published daily, weekly, monthly, quarterly or yearly; carry **volume, number and date** (e.g. vol. 60 (2) 2016).

Examples: journals, magazines, annuals, newspapers.

- **Non-book materials** (audio-visual): things you see and/or hear, not printed to read like a book: computers, films, video cameras, televisions, radios, slides, maps, charts, photographs.

Personnel

Group	Who is in it
Core library personnel	Academic Librarians and the Para-professional staff (Library Officers)
Supportive personnel	Library attendants/assistants, porters, clerical officers, technical staff of different categories

The four components are of **equal importance** and cannot be separated. Their roles are **complementary** (symbiotic): one cannot work well without the others.

> ⚡ **EXAM HACK**
>
> Four components = **B-B-P-C** (Books, Building, Personnel, Clientele). If an option lists only three of them, it is incomplete. Core staff = Librarians + Library Officers. Supportive staff = attendants, porters, clerks, technicians.

1.5 How a library builds its stock (acquisition methods)

★ ALZHEIMER'S MODE (the simple version)

How can a shop get goods? Buy them, receive gifts, swap with another shop, or inherit them. A library does the same four things: **Purchase, Gift, Exchange, Bequeath**.

Method	Meaning	Note
Purchase	Also called "**cash and carry**". Librarian visits the vendor, or invites the bookseller to the library. Some vendors **hawk** (go from institution to institution).	Costs money
Gift / Donation	Philanthropists, authors, institutions donate books and learning resources, willingly or when asked.	Comes free
Exchange	Swapping unwanted gifts or duplicates with other (sister) institutions for materials that are needed. Gifts are not normally rejected.	Swap, not buy

Method	Meaning	Note
Bequeath (Will)	A person states in his **will** that his books go to an institution after death.	Example: Chief Obafemi Awolowo left his law collection to Ogun State University (now Olabisi Onabanjo University); took effect when he died in 1987

⚡ **EXAM HACK**

Acquisition = **P-G-E-B**. Cash and carry = purchase. Will = bequeath. Swapping with sister institutions = exchange. If a question says “after his death”, answer bequeath.

1.6 Library rules and regulations

Where there is no law there is no offence, so users must know the rules. Offenders can be sanctioned. There are 21 rules; learn them in these groups:

- **Entry:** must be registered; registered users get a **library identity card** which gives access; show the card at the entrance; show books and papers you carry out to officials at the control point.

- **Behaviour:** keep silence in and around the library; orderly conduct everywhere; no seat reservation for others; no pets; no indecent dressing; no marking or defacing of books.
- **Prohibited things:** naked light; smoking, eating or drinking; **use and charging of cell phones**.
- **Borrowing:** follow due procedure for loans; return on or before the date due; keeping books beyond the date is an offence that attracts a fine; materials in **reference, serials or reserved sections** cannot be borrowed out.
- **Reading area:** leave consulted books on the reading table; never shelve a book yourself.
- **Security:** a suspected security risk may be ordered out; staff-only, private or reserved areas are off-limits without permission.

Sanctions for breaking the rules

Penalty depends on how serious the offence is. Serious offenders appear before the **Library Disciplinary Committee**, which recommends action as a warning to others. Sanctions range from:

- A **letter of undertaking** to be of good behaviour.
- Replacement of a lost book, or **payment in lieu**.
- **Suspension** from library use for a period.

- **Dismissal** from the university after trial and being found guilty.

> ⚡ **EXAM HACK**
>
> Sanctions go from mild to heavy: **Letter → Replace/pay → Suspend → Dismiss**. Never rank them any other way. Reference, serials and reserve materials never leave the library.

1.7 Chapter 1 recap

- Library from Latin *liber* = book.
- 6 types; 5 attributes; 4 components; 4 acquisition methods.
- Book = at least 49 pages; fewer = pamphlet.
- Public library = people's university; National library = legal deposit + ISBN/ISSN.
- Core staff = Librarians + Library Officers. Awolowo bequeathed his law books in 1987.

Exam hacks for this chapter (from the CBT)

> ⚡ **EXAM HACK**
>
> Types of library = APN-PRS: Academic, National, Private, Public, Research/Special, School. Six, never seven.

⚡ **EXAM HACK**

Four components = B-B-P-C: Books/materials, Building, Personnel, Clientele. If an option names only 3 of them, it is incomplete.

⚡ **EXAM HACK**

Four ways to acquire stock = P-G-E-B: Purchase, Gift, Exchange, Bequeath. Gift comes free, Purchase is cash and carry.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **The word 'library' comes from the Latin word 'liber'. What does 'liber' mean?**

   A. Shelf
   B. Reading
   C. Book
   D. Knowledge

   ✓ **Answer: C.** Library is from Latin 'liber' meaning book.

2. FUOYE Library and EKSU Library are examples of which type of library?

A. Academic library
B. School library
C. Special library
D. Public library

✓ **Answer: A.** Libraries attached to universities, polytechnics and colleges are academic libraries.

3. Which library has the legal deposit right and can issue ISBN and ISSN?

A. Academic library
B. Public library
C. Private library
D. National library

✓ **Answer: D.** The apex library of a nation has legal deposit and is empowered to issue ISBN and ISSN.

4. What does legal deposit right mean for a national library?

A. It gets government money for each book it lends out
B. It can borrow books from university libraries at no cost
C. It can receive copies of all works on the country
D. It can stop an author from publishing a book abroad

✓ **Answer: C.** Legal deposit = the right to acquire materials published on the country, for retention in its stock.

5. A lawyer who keeps law books at home for his own use has which type of library?

A. Public library
B. Academic library
C. Private library
D. School library

✓ **Answer: C.** Private libraries are owned by an individual or an organisation like a church.

6. Which library is called 'the people's university' and is paid for with tax-payers' money?

A. Public library
B. National library
C. Research library
D. Academic library

✓ **Answer: A.** Public libraries serve everybody in the community without discrimination, children included.

7. IITA Ibadan and the Central Bank of Nigeria have libraries of what type?

A. Special libraries
B. Academic libraries
C. School libraries
D. Public libraries

✓ **Answer: A.** Special libraries serve a defined group of specialist users, e.g. doctors or engineers.

8. Who should be in charge of a real library according to its attributes?

A. A person trained formally in librarianship
B. A teacher who has many books at home
C. The founder of the collection, whatever his training
D. Any educated person picked by the school head

✓ **Answer: A.** A librarian, library officer or teacher-librarian. A collection run by an untrained person is not a library.

9. Why is a room full of books, locked up and not open to users, not a library?

A. Because books must be replaced every year before use
B. Because a library must always have more than one thousand books
C. Because only a government building can be called a library
D. Because it keeps the collection out of users' reach

✓ **Answer: D.** People must be allowed to use the library at specified hours.

10. Which of these is NOT one of the four components of a library?

A. Clientele (users)
B. Funding department
C. Building or room
D. Personnel (staff)

✓ **Answer: B.** The four are books/serials/non-book materials, building, personnel and clientele.

11. What is a printed work of less than 49 pages called?

A. Manuscript
B. Pamphlet
C. Periodical
D. Serial

✔ **Answer: B.** Pamphlet = fewer than 49 pages.

12. Which of these describes serials?

A. Books printed once and kept in the reserve section only
B. Parts issued one after the other without end
C. Rare handwritten documents kept in the library archives
D. Films and slides that users can only see and hear

✔ **Answer: B.** Serials carry volume, number and date, e.g. journals, magazines, annuals.

13. Non-book materials are otherwise called what?

A. Directional materials
B. Reference materials
C. Printed materials
D. Audio-visual materials

✔ **Answer: D.** They are things that can be seen and heard, like films, slides, radios and computers.

14. Which of these is a non-book material?

A. Pamphlets
B. Slides
C. Annuals
D. Journals

✔ **Answer: B.** Computers, films, video cameras, TVs, radios, slides, maps, charts and photographs are non-book materials.

15. Which of these belong to the supportive library personnel?

A. The University Librarian
B. Academic librarians
C. Porters and attendants
D. Library officers

✔ **Answer: C.** Supportive staff: attendants/assistants, porters, clerical officers and technical staff.

16. Which of these is a sanction for breaking library rules?

A. A lifetime ban from all libraries in Nigeria
B. Withdrawal of the student's hostel space
C. Suspension from library use for some period
D. Loss of borrowing rights in another faculty library

✔ **Answer: C.** Sanctions: letter of undertaking, replacing the lost book or paying for it, suspension, or dismissal after trial.

17. The four components of a library cannot be separated and work together. This relationship is called what?

A. Competitive
B. Independent
C. Parasitic
D. Symbiotic

✔ **Answer: D.** Their roles complement each other, so they are symbiotic.

18. Books given to the library free by an author or an organisation came through which method?

A. Gift
B. Exchange
C. Purchase
D. Legal deposit

✔ **Answer: A.** Gift/donation materials come free; they are not bought.

19. A library receives a gift outside its curriculum and swaps it with a sister institution. What is this called?

A. Donation
B. Bequeath
C. Exchange
D. Purchase

✔ **Answer: C.** Exchange: trading unwanted gifts for materials the library actually needs.

20. Chief Obafemi Awolowo left his law collection to Ogun State University (now OOU) in his will. What method is this?

A. Purchase
B. Bequeath
C. Exchange
D. Gift

✔ **Answer: B.** A bequest comes into effect after the person dies. Awolowo died in 1987.

CHAPTER 2

Serials, Reference Sources & Intellectual Property Rights

2.1 Serials and periodicals

★ ALZHEIMER'S MODE (the simple version)

A serial is like a TV series: it keeps coming in parts and nobody has said when it will end. A periodical is a serial that comes out on a **regular schedule shorter than a year**, like a daily newspaper or monthly magazine.

KEY DEFINITION

Serials — vital information sources kept in the serial section. They are published in discrete parts, at regular or irregular intervals, with numerical or chronological designation, intended to continue **indefinitely**.

KEY DEFINITION

Periodicals — serials published at regular stated intervals of **less than a year** (daily, monthly, quarterly…). Examples: journals, magazines, newspapers.

- Serials have a title, volume, issue number or date, and an **ISSN** (International Standard Serial Number).
- Their information is current and up to date. Serials are **consulted only inside the library** and cannot be borrowed.

Type of serial	What it is	Examples
Journals	Original research reports or recent developments in a discipline. Articles go to editors, then to **peer review** by experts before acceptance. Published quarterly, bimonthly, monthly.	African Journal of Library, Archives and Information Science (twice a year)
Magazines	Articles on different topics; concise, informal, not necessarily written by experts; use graphics to attract readers.	Tell, Newswatch, Ovation
Conference proceedings	Records of a conference; often the full text of papers presented (not all conferences publish them).	—
Newspapers	News of current events, adverts, people's opinions.	Vanguard, Tribune
Newsletters	Smaller than newspapers; news about a particular organisation or activity.	—

Type of serial	What it is	Examples
Diaries	Usually handwritten account of events in a person's life; scientists note observations so others can repeat experiments.	—

⚡ EXAM HACK

Journal = **peer-reviewed research**. Magazine = **easy reading with pictures**. Every journal is a periodical and a serial; not every serial is a periodical. Serial = ISSN; Book = ISBN.

2.2 Reference sources

KEY DEFINITION

Reference sources — sources consulted for **quick, specific information** on a topic, theme, fact, date or word. They give direct information or direct you to where to find it; they are **not read from cover to cover** and are **not loaned out**. In the library they carry "**R**" or "**Ref**" on the spine.

There are two kinds:

Kind	Meaning	Examples
Directional	They **point** you to where the information can be found.	Bibliographies, indexes, abstracts

Kind	Meaning	Examples
Non-directional	They **give the information itself** directly.	Dictionary, encyclopaedia, directory, thesaurus, gazette, gazetteer, yearbook, almanac, handbook, map/atlas, manual, biography

Directional reference materials

- **Bibliography**: a list of published works (books, articles) by an author, on a subject, from a country or in a time frame, with bibliographic details (author, year, title, edition, journal title, place, publisher, pagination). Example: *Bibliography of Science and Technology*.
- **Indexes** contain keywords or index entries. **Abstracts** contain **summaries** of works so a researcher can decide whether the full text is needed. Examples: LISA (Library and Information Science Abstract), Dissertation Abstract International, Social Sciences Index, Science Citation Index.

Non-directional reference materials (the one-liners)

Source	What it gives you	Example
Dictionary	Alphabetical words with meanings, spellings, pronunciation, usage, origin (etymology), synonyms, antonyms. General or subject.	Advanced Learner's Dictionary; Dictionary of Medicine; Dictionary of Music
Encyclopaedia	Essay-like summaries on all aspects of knowledge, alphabetical, by experts, indexes at the end of the last volume. General or subject.	Britannica, Americana; Encyclopaedia of Architecture
Directory	Names of people, institutions, companies with addresses, phone numbers, affiliations, in classified or alphabetical order.	Nigerian Telephone Directory
Thesaurus	Words with **synonyms and antonyms**. Does **not** give meanings.	Roget's Thesaurus
Gazette	**Official government publication**: constitution, appointments, Acts, Bills, regulations, tenders.	Nigerian Gazette
Gazetteer	**Geographic names** of places and their features (rivers, cities, lakes, mountains).	Chamber's World Gazetteer
Yearbook	**Annual** publication: statistics, photos and facts about events of the **previous year**.	Commonwealth Universities Yearbook

Source	What it gives you	Example
Almanac	Dates, facts and statistics (economic, political, demographic), annual, in tables.	World Almanac and Book of Facts
Handbook	Brief factual information on one subject; handy and easy to reach.	CRC Handbook of Chemistry and Physics
Map / Atlas	Maps show the earth's surface or parts of it; an **atlas** is a collection of maps.	World Atlas

⚡ **EXAM HACK**

Pairs people mix up: **Gazette** (government news) vs **Gazetteer** (place names). **Thesaurus** (similar words, no meanings) vs **Dictionary** (meanings). **Yearbook** (last year's events) vs **Almanac** (dates and statistics in tables). Directional = D-I-A → Bibliographies, Indexes, Abstracts only.

2.3 Reference services

Reference services are all the activities of the **reference librarian** to meet a patron's information need:

- Answering enquiries and guiding users to resources; searching for information for patrons.
- Helping users **assess the quality** of a source.

- **Referral services** (sending a user to another library or resource); developing the reference collection.
- **User education**: introducing new users to sources, locations, services, building layout and rules.
- Compiling bibliographies; **Current Awareness Service (CAS)** and **Selective Dissemination of Information (SDI)**.
- Can be delivered by email, mailing list, telephone, online chat.

A reference librarian needs: **reference books**; a **computer with internet**, databases and e-resources; and **printed lists of holdings** not in the catalogue (e.g. newspapers).

2.4 Intellectual Property Rights (IPR)

> **★ ALZHEIMER'S MODE (the simple version)**
>
> Your idea belongs to you the way your phone belongs to you. If someone copies your song and sells it, they have stolen something you cannot touch but you own. The law that protects that is **Intellectual Property Rights**.

Intellectual property products are "the creative ideas and expressions of the human mind". They are **intangible** (you cannot touch them) but have value, so the law protects them. IP applies to **trademarks,**

patents and copyright.

Term	Meaning
Trademark	Symbols or signs put on products to show their **commercial origin**.
Patent	A legal document giving an **inventor** the right to use and profit from his invention for a **specified period**.
Royalties	Money paid to owners of creative works **each time the work is sold or performed**.
Copyright	The right given by law to owners of creative works to **monopolise the use** of their works or control its use by others. Symbol: ©. (the exclusive right given to an author and other creative artists to authorise the use of their work in any way.)

In Nigeria, copyright covers **five** categories: **literary works, musical works, artistic works, cinematograph films, sound recordings**.

> ⚡ **EXAM HACK**
>
> Nigerian copyright = **L-M-A-F-S**: Literary, Musical, Artistic, Films, Sound recordings. Trademark = the logo on a product. Patent = invention. Royalty = money per sale. Copyright = the © on creative work.

Exceptions to copyright: fair use and fair dealing

- **Fair use**: opportunity to cite or use a copyrighted work **without asking permission** (research,

criticism, news reporting, teaching, commentary). Not for commercial purposes; use only a limited portion that will not hurt the work's market.

- **Fair dealing**: allows reproduction of limited portions for **review, research or private study**.
- REPRONIG (2004): copyright is not infringed when a work is copied for **research or private study**; criticism, review or reporting of current events with acknowledgement; reasonable extracts with acknowledgement; judicial proceedings; or where it is shown to the public.

Violation of copyright

Violation	Meaning
Plagiarism	Copying another person's intellectual creation **without acknowledging the owner**. Also called academic dishonesty, a grievous offence. Even reusing your **own** previous work without acknowledgement is self-plagiarism.
Piracy	Making copies of copyrighted works **for financial gain**. It deprives the owner of income and discourages creativity.
Counterfeits	Reproducing **fake copies** of an original work for sale.
Photocopying	Allowed only for research or private study, and reasonably: a small part of the work, not enough to hurt the author's income.

⚡ **EXAM HACK**

Plagiarism = copying without **credit**. Piracy = copying for **money**. Counterfeit = **fake** copies. Quote the idea, give credit, and you are safe.

2.5 Referencing basics

KEY DEFINITION

Referencing (citing) — giving relevant, accurate and complete information that acknowledges the sources you used in an assignment, thesis, research article or report.

- It shows you researched widely, that the work is not only your own view, and it lets others **trace** the original works.
- You can **quote** directly (verbatim) or **paraphrase** (say the same idea in entirely different words). Either way, cite the owner in the text and in a reference list at the end.
- **Short quotation** goes in quotation marks; **long quotation** is indented.
- **Reference list**: every source you actually cited, usually alphabetical (sometimes numbered).
- **Bibliography**: can be the same as a reference list, or a list of sources you consulted but did not cite.

- **In-text citation**: author's surname and year, e.g. According to Alaba (2007)...
- **Footnotes** appear at the bottom of the page (small number in text points to it); **endnotes** appear at the end of the document. Not every style uses them.

Which style for which field?

Style	Used mostly in
Harvard	Social and behavioural sciences, philosophy, politics, economics, arts and **health sciences**
APA (American Psychological Association)	Social sciences, education, history
MLA (Modern Language Association)	Literature, arts, humanities
Chicago Manual of Style	Social sciences, history, political studies, theology, computer science
Turabian	A **simplified Chicago**, developed by Kate Turabian for theses, dissertations and term papers

⚡ **EXAM HACK**

Harvard = **H**ealth (also arts, economics). APA = **P**sychology/education. MLA = **L**iterature. Turabian = Chicago made easy. The style you use depends on the faculty, publisher or discipline.

Exam hacks for this chapter (from the CBT)

⚡ EXAM HACK

Directional = DIRECTS you elsewhere = Bibliographies, Indexes, Abstracts (B-I-A). Everything else (dictionary, atlas, almanac...) gives the answer directly.

⚡ EXAM HACK

Thesaurus gives words that mean the same or opposite but NEVER meanings. Dictionary gives meanings. Don't swap them.

⚡ EXAM HACK

Plagiarism = copying ideas without credit. Piracy = copying for money. Counterfeit = fake copy of the original. Money in the picture means piracy.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **Which of these is covered by copyright in Nigeria?**

   A. Names of products
   B. Scientific theories
   C. Machines and tools
   D. Sound recordings

   ✔ **Answer: D.** Literary, musical, artistic works, cinematograph films and sound recordings.

2. **Which of these is a reference service offered in the library?**

   A. SDI (Selective Dissemination)
   B. Cataloguing of newly bought books
   C. Stamping of newly acquired books
   D. Binding of damaged books for users

   ✔ **Answer: A.** Others: CAS, user education, referral services, compiling bibliographies and answering enquiries.

3. **What do we call serials published at regular intervals of less than a year?**

   A. Annuals
   B. Gazettes
   C. Periodicals
   D. Pamphlets

   ✔ **Answer: C.** Daily, weekly, monthly or quarterly serials are periodicals.

4. **Which serial carries original research reports that experts review before they are published?**

   A. Newspaper
   B. Magazine
   C. Journal
   D. Newsletter

   ✔ **Answer: C.** Editors send submitted manuscripts to reviewers before accepting or rejecting them.

5. **Tell, Newswatch and Ovation are examples of what?**

   A. Journals
   B. Magazines
   C. Gazettes
   D. Newspapers

   ✔ **Answer: B.** Magazines are concise, informal articles on different topics, often with graphics.

6. **Someone copies another person's ideas into his own work without giving credit. What is this offence called?**

   A. Counterfeiting
   B. Plagiarism
   C. Fair dealing
   D. Piracy

   ✔ **Answer: B.** Plagiarism = academic dishonesty. Even your own earlier work must be acknowledged.

7. Making copies of a copyrighted work in order to sell and make money is known as what?

A. Paraphrasing
B. Plagiarism
C. Piracy
D. Referencing

✔ **Answer: C.** Piracy takes away the owner's income and discourages him from producing more works.

8. How are reference books usually marked in the library?

A. With the ISBN on the date slip
B. With 'R' or 'Ref' on the spine
C. With a red sticker on the cover
D. With the word 'Serial' on the first page

✔ **Answer: B.** Reference books carry R or Ref on the spine and are not loaned out.

9. Making fake copies of an original work for sale is called what?

A. Plagiarism
B. Piracy
C. Counterfeiting
D. Citation

✔ **Answer: C.** Counterfeits are reproductions of fake copies of the original.

10. A reference source with words in alphabetical order and their meanings, spellings and usage is called what?

A. Directory
B. Thesaurus
C. Dictionary
D. Gazetteer

✔ **Answer: C.** A dictionary gives meaning, spelling, pronunciation and usage of words.

11. Roget's Thesaurus is used mainly to find what?

A. Meanings and pronunciation of words
B. Synonyms and antonyms of words
C. Names of places and where they are
D. Names and phone numbers of companies

✔ **Answer: B.** Thesauri list related words but do not give meanings.

12. Which source lists names of people or organisations with their addresses and phone numbers?

A. Almanac
B. Gazetteer
C. Directory
D. Atlas

✔ **Answer: C.** Directories list names, addresses, phone numbers and affiliations, e.g. the Nigerian Telephone Directory.

13. A gazetteer is used to find what?

A. Geographic names of places and their locations
B. Facts on one subject in short, handy form
C. Statistics and photographs of last year's events
D. Official notices issued by the government

✔ **Answer: A.** Chambers World Gazetteer is an example. Gazette (government) is different from gazetteer (places).

14. Which reference source is a yearly collection of dates, facts and statistics, often in tabular form?

A. Almanac
B. Yearbook
C. Atlas
D. Directory

✔ **Answer: A.** Almanacs are rich in economic, political and demographic information.

15. What is the difference between a short quotation and a long quotation?

A. A short one is italicised and a long one is underlined
B. A short one needs a footnote and a long one needs none
C. A short one goes in quotation marks and a long one is indented
D. A short one is indented and a long one goes in quotation marks

✔ **Answer: C.** Short = quotation marks. Long = indented block.

16. Short notes at the bottom of a page that point to a source or give more information are called what?

A. Abstracts
B. Prefaces
C. Endnotes
D. Footnotes

✔ **Answer: D.** Endnotes do the same job but come at the end of the document.

17. What is a patent?

A. An inventor's legal right to benefit from his invention
B. The right to copy any work for school use
C. A sign put on a product to show where it was made
D. Money paid each time a creative work is sold or performed

✔ **Answer: A.** Patent = inventor's right to use and make money from the invention for a set time.

18. Symbols or signs put on products to show their commercial origin are called what?

A. Copyrights
B. Patents
C. Trademarks
D. Royalties

✔ **Answer: C.** Think brand logo.

19. Which referencing style is a simplified version of the Chicago Manual of Style?

A. Harvard
B. MLA
C. APA
D. Turabian

✓ **Answer: D.** Kate Turabian developed it for theses, dissertations and term papers.

20. A student quotes a few lines from a book for research and acknowledges the author. This falls under what?

A. Plagiarism
B. Piracy
C. Counterfeiting
D. Fair use or fair dealing

✓ **Answer: D.** Fair use needs a limited portion, no commercial purpose, and no harm to the work's market.

CHAPTER 3

Book Anatomy, Citation & Referencing

3.1 What is a book?

KEY DEFINITION

Book — a medium for recording information as writing or images, usually of many pages (**not less than 49**), bound together and protected by a cover.

- The technical name for the physical arrangement of a book is **codex**. One sheet in a codex is a **leaf**; each side of a leaf is a **page**.
- **E-book**: text or illustrated pages in electronic format.
- A book has two major parts: the **preliminary part** and the **body of the text**.

3.2 Parts of a book (bibliographic parts)

★ ALZHEIMER'S MODE (the simple version)

Open a textbook slowly from the outside in: cover, then a blank sheet, then title pages, then who owns the book, then the contents, then the real text. That is exactly the order the parts come in.

Part	What it is
Cover	Every book has two covers (first and last). **Hard-bound** books are costlier but last longer; **soft-bound** are called **paperback**.
Spine	The edge where pages are **sewn or glued together**. Title and author's name are printed on it.
Dust jacket	A wrapper that protects the cover of a hard-bound book.
End paper	The blank sheet that comes before the preliminary pages.
Preliminary pages	Pages inside the front cover, before the main text. Unnumbered or numbered in **Roman figures** (not Arabic).
Half title	Gives less information than the full title page.
Title page	Where you find the **author's name, title, publisher, place and date of publication**.
Verso	The **back** of the title page. Gives copyright dates, copyright owner and other information.
Recto	The **right-hand** page of a book, usually with an odd page number. (Some notes misspell it "Rector". The right word is Recto.)
Copyright (©)	Statement that the contents belong to the author or publisher.

Part	What it is
Author / Publisher	The person who writes the book / the company that prepares and releases it to the public.
Year and place of publication	When, and in which country or town, the book was published.
ISBN	**International Standard Book Number**: a **13-digit** number given by the publisher to a book so it is distinguished from every other book. One ISBN belongs to one title in the whole world.
Preface	Where the author discusses briefly **all aspects of the book**.
Table of contents	Shows how the book is **divided into chapters**.
The text	The main part of the book.
Dedication	The author's way of showing **gratitude or appreciation to someone**.
Acknowledgement	Thanks to people who gave help or assistance during the preparation of the book.

⚡ EXAM HACK

Verso = back of the title page. Recto = right-hand, odd number. Preliminary pages use **Roman** numerals. ISBN has **13** digits and is for **books**; ISSN is for **serials**. Dedication = thanks to a person; Acknowledgement = thanks for help.

3.3 Why we reference and compile bibliographies

- To show you read widely and give credit to the real authors.
- To avoid plagiarism.
- To let readers trace the original works.
- To show your work is backed by other authors' views.
- To tell your reader where more information can be found.

> ⚠ **TRAP ALERT**
> **Reference** = only sources you actually cited in the text. **Bibliography** = may also include works you read but did not cite.

3.4 Harvard citation style

Harvard is also called the **author-date** method: in the text you insert the author's surname and the year in brackets. The references go in an alphabetical **list at the end**.

- In text: omit initials: **Alaba (2007)** at the start of a sentence, or.
- Same author, same year: add lowercase letters: **Alaba 2007a, Alaba 2007b, Alaba 2007c**.

- List alphabetically by first author, whatever the format. No need to separate books, journals, etc.
- More than one reference at the same place: list in **chronological order**, earliest first: Alokun (2000), John (2002) and Ajayi (2006).

Harvard layouts

Source	Layout and example
Book	Surname, initials (date). *Title: Subtitle.* Edition. Place: Publisher. Olajide, O. (2010). *A Perfect Guide to Basic Library Skills for Tertiary Institutions.* Lagos: Taisoft Publishers.
Journal	Surname, initials (date). Article title. *Journal title*, volume (part), pages.
2–3 authors	Blair G M, Jones R S and Simpson R H (1975). *Educational Psychology.* 4th ed. New York, Wiley.
4 or more authors	Use the first author and **et al.**: Stone G M et al (2006). *Principles of Biology.* 2nd ed. New York, Macmillan.
Edited book	Akinrinade S. et al (ed) (2004), *Locating the local in the global...* Ile-Ife, Obafemi Awolowo University.
No author / editor	Use the **name of the organisation** responsible: National Planning Commission (2004). *National Economic Empowerment and Development Strategy.* Lagos, NPC.

Source	Layout and example
Chapter in a book	Adefolalu, A.D. (2002). Climate. **In** Dominique M. et al (ed.) (2002). *Atlas of Nigeria.* Paris, Les Editions J.A.
Government publication	Nigeria Federal Ministry of Education (1990). *Blue print and action plan for the eradication of mass illiteracy by the year 2000.* Lagos, Macmillan.
Reference books	Cited by **title** unless there is an editor/compiler: *Oxford Advanced Dictionary* (1995). Oxford, Oxford University Press.
Thesis / dissertation	Must state the **level of degree** and the **awarding institution**.
Conference paper	Author (year). Title. **Paper presented at** the conference name, date, venue.
Newspaper / magazine	Include the **exact date** of publication: Nejo, A. (2019, March 25). Title. *The Punch Newspaper.* Retrieved from URL.
Website	Treated like printed sources, with the **Retrieved from URL**: Raosoft Inc. (2004). *Sample size calculator.* Retrieved from http://www.raosoft.com/samplesize.html

3.5 APA citation style

APA needs **two elements**: **reference citations in the text** and a **reference list**. In-text: author's last name and year in brackets, as in Harvard: Olajide (2020) asserts... or. Everything cited in the text must appear in the reference list.

- **Book**: Author. (Year). *Title.* Place: Publisher.
- **Journal**: Author, A. (year). Article title. *Journal Title, volume*(issue), pages.
- **2–3 authors**: separate with commas and an ampersand: Blair, G.M., Jones, B.S. & Simpson, R.H. (1975).
- **Chapter in a book** uses **(Eds.)** and **(pp. 62–65)**: Adefolalu, A.D. (2002). Climate. In M. Dominique et al. (Eds.), *Atlas of Nigeria* (pp. 62–65). Paris: Les Editions J.A.
- **Electronic sources**: link as close as possible to the exact document (not the home page) and provide **addresses that work**; give a title, a date (publication, update or retrieval) and a **URL**.
- Reports and theses: author, date, title, place, publisher; for a thesis add degree level and institution.

3.6 MLA citation style

MLA (Modern Language Association) documents sources with a **short parenthetical citation in the text** and a full entry in a list. The in-text citation has the **author's last name and the page number**, with no year: **(Olajide, 85)**.

- **Book**: Olajide, Olabode. *A Perfect Guide to Basic Library Skills for Tertiary Institutions.* Lagos: Taisoft

Publishers, 2010. (Title in quotes or italics, then publisher, then **year at the end**.)

- **Journal article**: Kalu-Uche, N. ... Journal name, volume, issue (year): page range. Note the year is in brackets near the **end**.
- **Chapter**: Adefolalu, Dayo A. “Climate”. *Atlas of Nigeria.* Ed. Dominique Mataillet et al. Paris: Les Editions J.A., 2002. 62–65.
- **Encyclopaedia article** and **Website** entries follow the same MLA pattern; websites include the posting date, the date you visited and the URL.

Feature	Harvard	APA	MLA
In-text			(Olajide, 85) — author + **page**
Year position in list	Right after author	Right after author	**At the end**
Author name	Surname + initials	Surname + initials	Surname, **full first name**

⚡ EXAM HACK

Quick style spotter: year right after the author = **Harvard or APA**. Year at the very **end** + page number in text = **MLA**. Full first names (Olajide, Olabode) = MLA. 4 or more authors in Harvard = **et al.** No author = use the organisation. Chapter in a book = the word **“In”**.

Conclusion: no style is better than another. The one you use depends on your discipline, institution, faculty and department. Follow the chosen format strictly, and if in doubt ask your lecturer, supervisor or editor.

Exam hacks for this chapter (from the CBT)

⚡ **EXAM HACK**

Book has TWO major parts: Preliminary pages (before the text, roman numerals) and the Body/Text. Preface, dedication, acknowledgement, table of contents all sit in the preliminary part.

⚡ **EXAM HACK**

Verso = back of the title page (copyright details). Recto = right-hand page (odd number). V comes before R, back before right.

⚡ **EXAM HACK**

Harvard = author-date. MLA = author + PAGE number, no year in the bracket. APA = author + year. Page number in the bracket points to MLA.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. What is the technical term for the physical arrangement of a book?

A. Leaf
B. Codex
C. Verso
D. Spine

✔ **Answer: B.** A single sheet in a codex is a leaf, and each side of a leaf is a page.

2. A single sheet in a codex is called what?

A. Page
B. Recto
C. Folio
D. Leaf

✔ **Answer: D.** Each side of a leaf is a page, so a leaf has two pages.

3. A book is made up of how many major parts?

A. Two
B. Four
C. Five
D. Three

✔ **Answer: A.** The preliminary part and the body of the text.

4. Which part of a book carries the title and author's name where the pages are glued or sewn together?

A. Dust jacket
B. Verso
C. End paper
D. Spine

✔ **Answer: D.** The spine is where the pages are joined, and the title and author are printed there.

5. What is the dust jacket used for?

A. To carry the date due slip
B. To list other works by the author
C. To protect a hard bound cover
D. To hold the call number of the book

✔ **Answer: C.** Not every book has one.

6. How are the printed sheets in the preliminary pages usually numbered?

A. In the same style as the text
B. Unnumbered or in roman figures
C. In Arabic figures starting from one
D. With alphabets from A onwards

✔ **Answer: B.** Arabic numerals belong to the text proper.

7. Which page carries less information than the full title page?

A. Verso
B. Preface
C. Dedication
D. Half title

✔ **Answer: D.** Half title = shorter version of the title page.

8. What is the verso of a book?

A. Back of the title page, with copyright details
B. The page where the author thanks those who helped him
C. The blank sheet at the start of the book
D. The right-hand page which carries an odd page number

✔ **Answer: A.** Verso gives copyright dates, the owners and other information.

9. What is the right-hand page of a book that carries an odd page number called?

A. Spine
B. Recto
C. Leaf
D. Verso

✔ **Answer: B.** Right = recto. The back of the title page is the verso.

10. What does ISBN stand for?

A. Internal Standard Book Network
B. International Serial Book Number
C. International Standard Book Number
D. Indexed Standard Bibliographic Name

✔ **Answer: C.** ISBN identifies a single book title. ISSN is for serials.

11. Where does an author discuss briefly all the aspects of his book?

A. Dedication
B. Table of contents
C. Preface
D. Acknowledgement

✔ **Answer: C.** The preface is the author's informal introduction to the book.

12. A page showing how a book is divided into chapters is called what?

A. Half title
B. Contents page
C. Preface
D. Dedication

✔ **Answer: B.** Table of contents lists chapters and where they start.

13. Which part of a book thanks people who gave help during its preparation?

A. Preface
B. Acknowledgement
C. Dedication
D. Verso

✔ **Answer: B.** Dedication = gratitude to a person. Acknowledgement = thanks for help rendered.

14. Why is the Harvard system also called the author-date method?

A. The date of the book is the only detail needed
B. The author's name and page number appear in brackets
C. The list of sources is arranged by date and not by author
D. Author's name and date are put in the text

✔ **Answer: D.** Example: Alaba (2007) in the text, then a full entry in the reference list.

15. In the Harvard system, how is the reference list arranged?

A. By date of publication, starting from the latest
B. In the order the sources appear in the text
C. By type of source, with books before journals
D. Alphabetically by the first author

✔ **Answer: D.** No need to separate books, journals, etc.

16. In Harvard, how do you cite a book that has four or more authors?

A. Give the first author's surname followed by 'et al'
B. Use the publisher's name in place of the authors
C. Use only the title of the book
D. List every author's name in full

✓ **Answer: A.** Example: Stone G M et al (2006).

17. A government document has no named author or editor. Whose name is used for the citation?

A. The responsible organisation
B. The year only, with no name
C. The name of the printer
D. The librarian who holds the copy

✓ **Answer: A.** E.g. National Planning Commission (2004).

18. The APA style needs which two elements for citing outside sources?

A. A glossary and an index of names
B. Text citations and a reference list
C. Footnotes and a bibliography only
D. Endnotes and a list of abbreviations

✓ **Answer: B.** APA uses the author's last name and year of publication in brackets in the text.

19. What is the difference between a reference list and a bibliography?

A. A reference list has works cited only; a bibliography may add works consulted
B. A reference list is for books, while a bibliography is for journals only
C. They mean different things and never appear in the same document
D. A reference list holds works consulted, while a bibliography holds only works cited
✔ **Answer: A.** A bibliography can also list sources you consulted but did not cite.

20. Which of these is a reason for giving references in a piece of writing?

A. It removes the need to read the original sources
B. It makes the work longer to meet the word count
C. It lets others trace the original works you used
D. It shows the writer is more important than the authors
✔ **Answer: C.** Referencing also shows you researched widely and protects you from plagiarism.

CHAPTER 4

Catalogue, Classification & Sections of the Library

4.1 Why a library must be organised

★ ALZHEIMER'S MODE (the simple version)

Imagine a market where every trader drops goods anywhere. You could never find the pepper seller. Libraries hold thousands or millions of items, so they arrange everything and make **lists** that tell you where each item is. Those lists are called **retrieval tools**.

The retrieval tools are: bibliographies, catalogues, indexes, finding aids and registers.

Retrieval tool	What it does	Where it is used
Bibliography	List of document surrogates (books, articles, conference papers) arranged by subject, form, author or place. Each item is listed **once** only.	Libraries
Index	A guide to the **intellectual content** of a document, showing where an item can be found. Alphabetical by subject or name.	Books, journals

Retrieval tool	What it does	Where it is used
Finding aids	Also called **inventories**; list items by personal or corporate source, not by individual author.	**Archives**
Register	Contains the identity and **donor** of each object; each object gets an identification number.	**Museums**
Catalogue	List of all items in a collection, by author, title or subject. The index to a library's collection.	**Libraries** (the major tool)

⚡ **EXAM HACK**

Match place to tool: **Archives = Finding aids**, **Museums = Registers**, **Libraries = Catalogue**. A catalogue is the "key or pointer" to the library's holdings.

4.2 The library catalogue

KEY DEFINITION

Catalogue — the list of records of the documents held in a library. When it records the holdings of **more than one library** it is a **Union Catalogue**.

Importance of the catalogue

- It **enhances access** to library collections.
- Lets users find books on a particular **subject**.
- Helps identify sources of information.

- Lets you find books by a particular **author**.
- Gives **full bibliographic details** of a book.
- Tells you whether the library has a certain book.
- Shows at a glance **how many copies** the library holds of a title.

> **KEY DEFINITION**
>
> **Cataloguing** — describing a book or other library item so that all its bibliographic details are highlighted: author first, title, publisher, place and date of publication, **pagination** (number of pages) and whether it has illustrations. The person who does it is the **cataloguer**.

Types of catalogue

- **Card catalogue**
- **OPAC** (Online Public Access Catalogue, computer-based)
- **Book catalogue**
- **Shelf catalogue**
- **CD-ROM catalogue**
- **COM** (Computer Output Microform) catalogue

The **card catalogue** is the one most used in developing countries like Nigeria, while **OPAC** is common in developed countries.

	Card catalogue	OPAC
Form	Each record on a **3 inches by 5 inches** card, filed in drawers in a catalogue cabinet	Electronic; the **most modern** form
Cards per item	At least **three** cards: author, title and subject	One record reachable through several points
Strength	**Flexible**: old cards removed, new ones inserted	Flexible and easily updated; links to outside libraries and current journals and books the library may not own

Approaches (access points)

In an academic library the catalogue is a **divided catalogue** with three approaches: **Author, Title and Subject**.

> ⚡ **EXAM HACK**
>
> 3 x 5 card, 3 cards per item (A-T-S: Author, Title, Subject). Nigeria = card catalogue. Developed countries = OPAC. More than one library in one catalogue = Union catalogue.

4.3 Classification

★ ALZHEIMER'S MODE (the simple version)

Classification is putting books that talk about the same thing on the same shelf, like putting all drinks together and all rice together in a shop. Each book gets a number or letter code for its subject so it has an exact home.

KEY DEFINITION

Classification — grouping things (library materials) **according to subject** in a logical order so they can be found and used easily. The classification section is headed by the **Chief Cataloguer**.

Six known schemes: **Library of Congress (LC), Dewey Decimal (DDC), Colon, Bibliographic, Moys, Universal Decimal (UDC)**. In Nigeria: **LC is used by academic libraries**; **DDC is used by national and public libraries**.

Library of Congress (LC): uses letters

Letter	Subject	Letter	Subject
A	General works	L	Education
B	Philosophy, Psychology, Religion	M	Music
H	Social sciences (economics, sociology)	N	Fine arts
J	Political science	P	Language and literature

Letter	Subject	Letter	Subject
K	Law	Q	Science
R	Medicine	S	Agriculture
T	Technology	Z	Bibliography, Library science, Information resources

Sub-divisions: **AE** Encyclopaedia, **AG** Dictionaries, **AN** Newspapers, **AS** Academic and learned societies, **AY** Yearbooks, almanacs, directories, **BF** Psychology, **PN** Literature, **PS** American literature.

Dewey Decimal Classification (DDC): uses numbers

DDC divides all knowledge into **ten main classes**:

Number	Class	Number	Class
000–099	Generalities	500–599	Pure science
100–199	Philosophy and related	600–699	Technology
200–299	Religion	700–799	Arts
300–399	Social sciences	800–899	Literature
400–499	Languages	900–999	General geography and history

Science division: 500 general science, **510 Mathematics**, **520 Astronomy**, **530 Physics**, **540 Chemistry**, **550 Geology**, 560 Paleontology, 570 Anthropology/Biology, **580 Botany**, **590 Zoology**.

⚡ **EXAM HACK**

LC = **letters** (R = Medicine, Q = Science, K = Law, Z = Library science, T = Technology). DDC = **numbers** (500 Science, 600 Technology, 800 Literature). Science sequence 510-590: **M-A-P-C-G-P-A-B-Z** = Maths, Astronomy, Physics, Chemistry, Geology, Paleontology, Anthropology/Biology, Botany, Zoology. Medicine student tip: R in LC, 610 in DDC.

4.4 How books get to the shelf

After acquisition and verification, every item goes through **processing** in this order:

Step	What happens
1. **Stamping**	The institution's ownership stamp is placed (e.g. Federal University Oye-Ekiti Library).
2. **Accessioning**	A unique serial number, the **accession number** (e.g. 00045), is given as soon as the item arrives. It numbers materials serially, so **no two items share one**.
3. **Cataloguing and classifying**	Descriptive cataloguing: author, title, bibliographic details; then the classification number is assigned.
4. **Call mark**	Also called **location mark** or **shelf mark**. It tells where the item sits on the shelf. It is the **classification number + the Cutter number**.

Step	What happens
5. **Other processing**	Paste the typed call mark on the spine, the **date due slip**, the **book envelope**; insert the typed **book card**; take statistics; **shelve** appropriately.

Cutter number: an alphanumeric symbol (first letter of the author's name, or title or corporate body, plus numbers from the Cutter table) added to the classification number to arrange items **within one class**.

> ⚡ **EXAM HACK**
> Order = **S-A-C-C-P-S**: Stamp, Accession, Catalogue, Call mark, Paste, Shelve. Accession number = unique per item; call mark = class number + Cutter number.

4.5 Sections and departments of a university library

Department	Sections inside it and what they do
Administrative	Headed by the **University Librarian**, who coordinates all departments. Duties: develop library policies, coordinate systems, **train staff**, keep records and financial expenditure, regular statistics of books catalogued, supervise and organise the library.

Department	Sections inside it and what they do
Technical Services (“behind the scenes”)	**Acquisition** (selects and acquires materials, checks stock to avoid duplicates); **Cataloguing** (catalogues and classifies); **Serials** (selects, orders, receives, shelves journals, magazines, reports, newspapers, proceedings); **Reprography and Bindery** (photocopying and binding).
Readers’ Services	**Circulation** (loan and return, shelf reading, registering new users, overdue notices, fines); **Reserve** (materials in constant demand but short supply, not loaned out); **Reference** (reference and advisory service, materials consulted not borrowed).
Research, Documentation and Bibliographic	In academic and research libraries: documents, private collections, **maps and manuscripts**.
Non-Print Media Resources	Audio-visual materials: slides, transparencies, pictures, photographs and software with their hardware.

⚡ EXAM HACK

Technical = **behind the scenes** (A-C-S-R: Acquisition, Cataloguing, Serials, Reprography). Readers’ = **front of house** (C-R-R: Circulation, Reserve, Reference). Fines and overdue notices = Circulation. Heavy-demand but few copies = Reserve. Duplicates check = Acquisition.

Exam hacks for this chapter (from the CBT)

⚡ EXAM HACK

LC for Academic libraries, Dewey for National and Public libraries. Think: L-C = Learned Campus, Dewey = the people's library.

⚡ EXAM HACK

Dewey: 10 classes, each 100 numbers wide. 500s = pure science, 600s = technology. Science sub-order is M-A-P-C-G-P-A-B-Z: Maths 510, Astronomy 520, Physics 530, Chemistry 540, Geology 550... Zoology 590.

⚡ EXAM HACK

Technical Services = 'behind the scene' (Acquisition, Cataloguing, Serials, Reprography/Bindery). Readers' Services = where users meet staff (Circulation, Reserve, Reference).

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. What does an index do?

A. Shows where an item in the collection can be located
B. Records how many copies of a title the library has
C. Lists the donors of objects kept in a museum
D. Gives a full summary of every book in the library

✔ **Answer: A.** An index is usually arranged alphabetically by subject or name.

2. Which retrieval tool is used in archives?

A. Indexes
B. Registers
C. Catalogues
D. Finding aids

✔ **Answer: D.** Finding aids are also called inventories.

3. Which retrieval tool is used mainly in museums and carries an identification number for each object?

A. Register
B. Catalogue
C. Bibliography
D. Finding aid

✔ **Answer: A.** A register has the identity and donor of each museum object.

4. Which of these is a reason why a library catalogue is important?

A. It shows the price of every book the library bought
B. It lists the names of users who borrowed a book
C. It stores the books that are in constant demand
D. It tells users if a book is in the library

✓ **Answer: D.** It also shows the number of copies of a title and gives full bibliographic details.

5. What is the person who describes library items to produce catalogue entries called?

A. Circulation officer
B. Acquisition officer
C. Cataloguer
D. Classifier

✓ **Answer: C.** Cataloguing = describing an item so its bibliographic details are highlighted.

6. How many cards are made for each item in a card catalogue?

A. At least three: author, title and subject
B. Only one card with all the access points
C. Five cards, one for each library section
D. Two cards: author and publisher

✓ **Answer: A.** Author card, title card and subject card.

7. What is the main advantage of the card catalogue?

A. It can be used from outside the library over the internet
B. Flexibility, as cards can be added or removed
C. It needs no staff to prepare and file the entries
D. It can show the records of libraries around the world

✔ **Answer: B.** The cards can easily be interfiled.

8. A divided catalogue in a university library is divided into how many parts?

A. Two: author and subject
B. Three: author, title and subject
C. Four: author, title, subject and date
D. Five: one for each faculty

✔ **Answer: B.** The three approaches are Author, Title and Subject.

9. What does classification mean in the library context?

A. Arranging library materials by size and colour
B. Listing the books the library holds in a catalogue
C. Grouping materials by subject in logical order
D. Giving each book a unique serial number on arrival

✔ **Answer: C.** Without it a systematic library cannot be built.

10. Which classification scheme is used by national and public libraries in Nigeria?

A. Moys
B. Colon
C. Dewey Decimal
D. Library of Congress

✔ **Answer: C.** Dewey is for national and public libraries.

11. In the Library of Congress scheme, which letter stands for Medicine?

A. S
B. Q
C. R
D. M

✔ **Answer: C.** Q = Science, R = Medicine, S = Agriculture, T = Technology.

12. In the Library of Congress scheme, which letter stands for Law?

A. H
B. L
C. J
D. K

✔ **Answer: D.** J = Political science, K = Law, L = Education.

13. In Library of Congress, which letter stands for Science?

A. R
B. Q
C. S
D. T

✔ **Answer: B.** Q = Science.

14. How many main classes does the Dewey Decimal scheme have?

A. Eight
B. Twenty-six
C. Ten
D. Twelve

✔ **Answer: C.** 000 to 999, ten classes of 100 numbers each.

15. Under Dewey, which numbers cover Zoology?

A. 590 to 599
B. 540 to 549
C. 580 to 589
D. 570 to 579

✔ **Answer: A.** 580 = Botany, 590 = Zoology, 540 = Chemistry.

16. Under Dewey, which numbers cover Physics?

A. 530 to 539
B. 510 to 519
C. 520 to 529
D. 540 to 549

✔ **Answer: A.** 510 Mathematics, 520 Astronomy, 530 Physics, 540 Chemistry.

17. What is a call mark?

A. A stamp showing which library owns the material
B. A mark showing the day a borrowed material must be returned
C. A mark made of the class number and cutter number
D. A number given to the material by its publisher

✔ **Answer: C.** Also called location mark or call number.

18. What is a cutter number used for?

A. To count the number of books in a class
B. To show that a book is on loan
C. To identify the publisher of a book
D. To order items within a class

✔ **Answer: D.** It uses the first letter of the author's name and numbers from the cutter table.

19. Where are library materials charged (loaned out) and discharged (returned)?

A. Reference section
B. Circulation section
C. Reserve section
D. Serials section

✔ **Answer: B.** Circulation also registers new users, sends overdue notices and collects fines.

20. Materials that are in constant demand but short in supply are kept in which section?

A. Reprography section
B. Circulation section
C. Reserve section
D. Reference section

✔ **Answer: C.** They are given to users on request but not loaned out.

CHAPTER 5

Library Software & Electronic Databases

5.1 Why libraries use software and databases

Libraries now hold **printed and electronic** resources. Electronic documents can be stored, accessed and delivered as needed, so a library's service is no longer limited to its four walls. Libraries are becoming **virtual libraries** where books and journals have turned into **e-books and e-journals**. Software and databases are popular because of **portability, flexibility and instant availability of up-to-date information**.

5.2 Library software

★ ALZHEIMER'S MODE (the simple version)

Software is the set of instructions that tells the computer what to do, like a recipe tells a cook what to do. **System software** runs the machine itself; **application software** helps you do a job. **Library software** is an application for running a library.

Type	Meaning	Examples
System software	Controls the hardware and gives the basic functions of the computer. Includes device drivers, utilities and operating systems.	Windows, Linux, Unix, Mac OS, Android
Application software	Helps you do a task using the computer.	Microsoft Office, Corel Draw, Photoshop, browsers, library software

KEY DEFINITION

Library software — computer software packages designed to perform **library housekeeping functions**: managing, cataloguing and circulating materials to patrons. Functions: **cataloguing, circulation, acquisition, reference, serials management, web-based OPAC, stock management**.

Good library software gives management, control and easy access to resources and **reduces time wasted** serving users.

Two models of library software

	Proprietary	Open source
Meaning	You **pay a subscription fee**	Free software for library routines. Not always cost-free, but **free to use, modify and share**
Examples	Symphony, Millennium, Aleph, Voyager, Polaris, Carl X, Spydus	Koha, Evergreen, Greenstone, D-Space, NewGenLib, Emilda, OpenBiblio, PMB, PhpMyLibrary, Senayan...
Note	Mature and still the dominant approach for automation	**Koha** is the one used at **FUOYE**

⚡ **EXAM HACK**

Koha = Federal University Oye-Ekiti Library (open source). If you see Symphony, Millennium, Aleph or Voyager, think **proprietary**. Open source = free to **use, modify and share** (but not always free of cost).

5.3 Electronic databases

KEY DEFINITION

Electronic database — a collection of data or information specially organised for **rapid search and retrieval by a computer**. A library database is a searchable electronic index of published, reliable resources (journals, newspapers, magazines, sometimes e-books and multimedia) not freely available on Google. Valued for **speed, flexibility, wide range and currency**.

- Some databases are **free**; others need a **subscription fee**. Universities subscribe so lecturers and students can use them.

Database	What it is
AGORA	Access to Global Online Research in Agriculture, an initiative of the **Food and Agriculture Organization (FAO)** with publishers; about 2,900 institutions in 100+ countries.
EBSCOhost	Leading provider of research databases, e-journals, magazines, e-books and discovery service to libraries.
HINARI	Health InterNetwork Access to Research Initiative, by the **World Health Organization** and publishers: biomedical and health literature. **Your course!**
OARE	Online Access to Research in the Environment, led by the **UN Environment Programme (UNEP)**.

Database	What it is
DOAJ	Directory of Open Access Journals: indexes high-quality, peer-reviewed **open access** journals (10,000+).
ScienceDirect	Subscription-based access to scientific and medical research: 12 million items, 3,500 journals, 34,000 e-books.
PubMed	Free full-text archive of biomedical and life sciences journal literature at the **US National Library of Medicine, NIH**.
TEEAL	The Essential Electronic Agricultural Library: an **offline**, searchable digital library of agricultural journals.
ProQuest	Many databases: dissertations, theses, e-books, newspapers, periodicals, historical collections, government archives.

⚡ **EXAM HACK**

Agency memory: **AGORA = FAO** (agriculture), **HINARI = WHO** (health), **OARE = UNEP** (environment), **PubMed = NIH**. TEEAL = the only **offline** one. DOAJ = **open access**. ScienceDirect = **subscription**. As a health student, HINARI and PubMed are your friends.

5.4 Benefits and concern

- Easy access to information resources and to the **web-based OPAC**.

- Improved efficiency of library services; easy generation of records.
- Less time to serve users.
- **Concern:** many users do not know how to use these resources. Librarians must **train and guide** users, and users need basic **ICT skills**.

Exam hacks for this chapter (from the CBT)

⚡ EXAM HACK

Match the database to the first letter of its field: HINARI = Health (WHO), AGORA = Agriculture (FAO), OARE = Environment (UNEP), TEEAL = Agriculture (offline), DOAJ = Open access journals, PubMed = Biomedical.

⚡ EXAM HACK

Two software types: System (runs the machine: Windows, Linux, Android) and Application (does a job for the user: Word, Photoshop, library software). Library software is application software.

⚡ EXAM HACK

Proprietary = you pay (Symphony, Millennium, Aleph, Voyager). Open source = free to use, modify and share (Koha, Evergreen, Greenstone, DSpace). FUOYE runs Koha.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. Software is divided into how many main types?

A. Three: system, application and network software
B. Two: system software and application software
C. Four: input, output, process and storage software
D. Five: one for each unit of the computer

✓ **Answer: B.** System software and application software.

2. Which of these is system software?

A. Linux
B. Microsoft Word
C. Photoshop
D. Koha

✓ **Answer: A.** System software also covers device drivers, utilities and servers.

3. Which of these is application software?

A. Unix
B. Photoshop
C. Android
D. Linux

✓ **Answer: B.** Applications include Microsoft Office, Corel Draw, browsers and library software.

4. Which of these is a library housekeeping function that software can handle?

A. Circulation
B. Fumigation
C. Rubber stamping
D. Book binding

✔ **Answer: A.** Others: acquisition, cataloguing, serial processing, web-based OPAC and stock management.

5. Library software comes in how many models?

A. Five: one for each type of library
B. Two: proprietary and open source
C. Four: local, national, regional and global
D. Three: free, trial and paid

✔ **Answer: B.** Proprietary software and Open Source Software (OSS).

6. Which of these is a proprietary library software package?

A. Symphony
B. Koha
C. Greenstone
D. Evergreen

✔ **Answer: A.** Others: Millennium, Aleph, Voyager, Polaris, Carl X and Spydus.

7. Which of these is an open source library software?

A. Voyager
B. Koha
C. Millennium
D. Aleph

✓ **Answer: B.** Others: Evergreen, Greenstone, DSpace, NewGenLib and OpenBiblio.

8. Open source software is described as what?

A. Always free of charge and never needing money to run
B. Free to use, modify and share
C. Locked to one library and never to be changed
D. Meant for use only by foreign libraries

✓ **Answer: B.** Free refers to freedom to use, modify and share.

9. An electronic database is best described as what?

A. A printed list of all books borrowed by students in a year
B. A set of programs that control the hardware of a computer
C. A store of paper files kept in a library cabinet
D. An organised collection arranged for fast computer search

✓ **Answer: D.** It is a searchable electronic index of published, reliable resources.

10. AGORA is an electronic database in which field?

A. Agriculture
B. Health
C. Environment
D. Law

✔ **Answer: A.** AGORA = Access to Global Online Research in Agriculture, led by the FAO of the United Nations.

11. Which organisation leads the HINARI programme?

A. National Institutes of Health
B. United Nations Environment Programme
C. World Health Organization
D. Food and Agriculture Organization

✔ **Answer: C.** HINARI = Health InterNetwork Access to Research Initiative.

12. OARE gives developing countries access to research in which field?

A. Law
B. Agriculture
C. Environment
D. Health

✔ **Answer: C.** OARE = Online Access to Research in the Environment, led by UNEP.

13. Which database gives subscription-based access to a large body of scientific and medical research?

A. PubMed
B. TEEAL
C. DOAJ
D. ScienceDirect

✔ **Answer: D.** ScienceDirect hosts millions of pieces of content from thousands of journals and e-books.

14. PubMed is an archive of which kind of literature?

A. Newspapers and magazines
B. Agricultural reference journals
C. Environmental science research
D. Biomedical and life sciences

✔ **Answer: D.** It is a free full-text archive run by the National Library of Medicine, NIH.

15. EBSCOhost is mainly known as what?

A. A directory of agricultural journals kept offline
B. A program for running a library's circulation
C. A free archive run by the United Nations
D. A provider of research databases and e-books

✔ **Answer: D.** It also offers a discovery service to libraries.

16. Which database gives access to dissertations, theses, newspapers, periodicals and historical collections?

A. OARE
B. HINARI
C. TEEAL
D. ProQuest

✔ **Answer: D.** ProQuest is a collection of many databases.

17. What concern did the writer raise about e-resources in libraries?

A. They are too cheap to be taken seriously
B. They are only meant for foreign students
C. Many users do not know how to use them
D. They cannot be updated once bought

✔ **Answer: C.** Users need ICT skills, so librarians should train them.

18. Libraries are said to be turning into virtual libraries because of what?

A. Buildings are being sold off by universities
B. Students prefer not to visit the library at all
C. Books and journals are turning into e-books
D. Printed books are no longer allowed in libraries

✔ **Answer: C.** This has increased the global dissemination of information.

19. Who provides free databases to libraries in developing countries?

A. The local government
B. The library's own staff
C. Publishers or vendors
D. The students' union

✓ **Answer: C.** Some are free; others need a subscription fee.

20. Specialised electronic databases are said to hold published records that are not available on what?

A. The library's own shelves
B. Google or search engines
C. Printed journals and magazines
D. Any computer in the university

✓ **Answer: B.** Specialised records, often in full text.

CHAPTER 6

Sources of Information (Primary, Secondary & Tertiary)

6.1 What is information?

Information is difficult to define because it means different things to different people. Dictionary.com: **knowledge communicated or received concerning a fact or circumstance**. Oxford: **facts provided or learned about something or someone**; also **data processed, stored or transmitted by a computer**.

There are three basic sources: **primary, secondary and tertiary**. Which one you consult depends on the type of information you need.

> ★ **ALZHEIMER'S MODE (the simple version)**
> Think of a football match. **Primary** = you were in the stadium (first-hand). **Secondary** = a sports journalist explains what happened. **Tertiary** = a list that tells you which reports and videos exist.

6.2 Primary sources

Original materials on which other sources are based. They report a discovery or share new information, give **first-hand accounts**, and present information in its **original form, with no interpretation or evaluation** by others. Often the first formal appearance of research results.

Examples: diaries, letters, speeches, patents, photographs, **newspapers**, **journal articles**, theses and dissertations, survey research, proceedings of meetings and conferences, original documents (birth certificates, wills, marriage licences, trial transcripts), records of organisations and governments (treaties, annual reports), e-mail, interviews, video and audio recordings, websites, **artefacts** (coins, plant specimens, fossils, furniture, tools, clothing), **works of art** (paintings, sculptures, musical scores, buildings, novels, poems), architecture, literature, music.

6.3 Secondary sources

Created from primary sources. Their authors had **no first-hand experience** of the event. They describe, analyse, interpret, evaluate, comment or discuss the evidence from primary sources. They are **not evidence**, they are **commentary on evidence**.

Examples: **textbooks**, biographical works, commentaries, criticisms, **dictionaries**,

encyclopaedias. Always note **how recent** a secondary source is.

6.4 Tertiary sources

Pointers to primary and secondary sources: works that list primary and secondary resources in an area, or that index, organise and compile citations to secondary sources.

Examples: **chronologies, bibliographies, indexes, encyclopaedias, directories**.

	Primary	Secondary	Tertiary
Role	Original, first-hand	Interprets primary	Points to primary and secondary
Author was there?	Yes	No	Not relevant
Examples	Diary, patent, thesis, journal article, interview, artefact	Textbook, biography, commentary, criticism	Bibliography, index, chronology, directory

⚠ **TRAP ALERT**

Encyclopaedias and dictionaries appear as **secondary** in the usual lists, and **encyclopaedias** also appear in the **tertiary** list. In an exam, if only one of them is offered, pick the tertiary option only when the question says “pointer / list / index”. Journal articles, newspapers, theses and patents are **primary**.

⚡ **EXAM HACK**

P = **P**roof (original). S = **S**tory about the proof. T = **T**racker (points you to others). Bibliography, index, directory, chronology = tertiary. Textbook, biography, criticism = secondary. Diary, patent, thesis, artefact = primary.

Exam hacks for this chapter (from the CBT)

⚡ **EXAM HACK**

PRIMARY = first-hand, raw, original (diary, letter, speech, patent, thesis, interview, artefact). SECONDARY = someone talking ABOUT the original (textbook, commentary, criticism). TERTIARY = points you to the others (bibliography, index, directory, chronology).

⚡ **EXAM HACK**

Quick test: did the writer see it happen? Yes = primary. Did the writer read about it? = secondary. Does it just list where to look? = tertiary.

⚡ **EXAM HACK**

Textbook, dictionary and encyclopedia are secondary. Bibliography, index and directory are tertiary. Don't swap the two lists.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **What decides which source of information a person should consult?**

   A. The age of the person looking for it
   B. The price of the material
   C. The kind of information needed
   D. The size of the library the person is using

   ✓ **Answer: C.** The choice of source depends on the type of information needed.

2. What are primary sources of information?

A. Materials made from a mix of other books
B. Materials that explain and comment on other sources
C. Materials that point you to other sources
D. Original materials on which other sources are based

✔ **Answer: D.** They report a discovery or share new information, first-hand.

3. A scientist's handwritten record of daily observations in an experiment is what kind of source?

A. Reference source
B. Tertiary source
C. Primary source
D. Secondary source

✔ **Answer: C.** Diaries are first-hand records, so they are primary.

4. Which of these is a primary source?

A. Chronologies
B. Patents
C. Encyclopedia
D. Biographical works

✔ **Answer: B.** A patent is the original record of an invention.

5. Which of these is a primary source?

A. Indexes
B. Directories
C. Textbooks
D. Thesis

✔ **Answer: D.** Original research results appear in theses, dissertations and journal articles.

6. A coin or a fossil kept in a museum is what kind of source?

A. Tertiary source
B. Secondary source
C. Reference source only
D. Primary source

✔ **Answer: D.** Artefacts like coins, plant specimens, tools and furniture are primary.

7. A musical score, a painting and a novel are examples of what?

A. Works of art counted as primary sources
B. Indexes counted as tertiary sources
C. Textbooks counted as secondary sources
D. Directories counted as reference sources

✔ **Answer: A.** Paintings, sculptures, novels, poems and buildings are primary sources.

8. Birth certificates, wills and marriage licences are examples of what?

A. Primary sources
B. Tertiary sources
C. Directional sources
D. Secondary sources

✓ **Answer: A.** Original documents are primary.

9. What are secondary sources of information?

A. Sources written from primary sources by outsiders
B. Original records written by the person who saw the event
C. Lists that only point to books and articles
D. Raw data collected during experiments

✓ **Answer: A.** They describe, analyse, interpret, evaluate or discuss evidence from primary sources.

10. A student reads a history textbook about a civil war. What kind of source is the textbook?

A. Primary source
B. Artefact
C. Secondary source
D. Tertiary source

✓ **Answer: C.** The author did not take part in the war; he wrote from other sources.

11. Which of these is a secondary source?

A. Photographs
B. Letters
C. Original speeches
D. Biographical works

✔ **Answer: D.** The writer of a biography did not live the events first-hand.

12. Which of these is a secondary source?

A. Artefact
B. Bibliography
C. Criticism
D. Treaty

✔ **Answer: C.** Criticisms evaluate other works.

13. Why is a secondary source not counted as evidence?

A. It is always older than the primary source
B. It is commentary on the evidence
C. It is written without any research at all
D. It is found only in the reference section

✔ **Answer: B.** Secondary sources are commentary on evidence, not the evidence itself.

14. What are tertiary sources of information?

A. Personal letters and diaries of famous people
B. Works that point to primary and secondary sources
C. Original records of research results
D. Commentaries written by experts on primary sources

✔ **Answer: B.** They list, index, organise and compile citations of other sources.

15. Which of these is a tertiary source?

A. Bibliography
B. Criticism
C. Diary
D. Textbook

✔ **Answer: A.** Chronologies, bibliographies, indexes, encyclopedias and directories.

16. Which of these is a tertiary source?

A. Directory
B. Biography
C. Interview
D. Thesis

✔ **Answer: A.** Directories list people and organisations.

17. Which of these is a tertiary source?

A. Commentary
B. Chronology
C. Photograph
D. Dissertation

✔ **Answer: B.** A chronology lists events in order of time and points to the sources.

18. A researcher wants a list of all articles ever written on malaria. Where should he start?

A. A tertiary source such as a bibliography
B. An artefact such as a fossil
C. A primary source such as a diary
D. A secondary source such as a criticism

✔ **Answer: A.** Tertiary sources point you to primary and secondary ones.

19. Which pairing of source type and example is correct?

A. Secondary - patent
B. Primary - textbook
C. Primary - bibliography
D. Tertiary - directory

✔ **Answer: D.** Directory is tertiary; textbook is secondary; patent is primary; bibliography is tertiary.

20. How do authors of secondary sources differ from authors of primary sources?

A. They write only for newspapers and magazines
B. They did not witness the event first-hand
C. They are never allowed to cite other works
D. They always write in the same year the event happened

✔ **Answer: B.** Secondary authors work from the evidence that primary sources provide.

CHAPTER 7

Online Searching (Challenges, Strategies & Evaluation)

7.1 What is online searching?

★ ALZHEIMER'S MODE (the simple version)

Searching online is like asking a very fast librarian who has already made a giant index. You type words, it matches them to its index and shows the best matches first. It is **not** searching the whole internet live.

KEY DEFINITION

Online searching — the process of **interactively searching for and retrieving information using a computer from online databases**, using a particular string of words or characters entered in a search engine.

A **search engine** is software that lets you query its **index** and returns results ranked by **relevance**. It receives your request, compares it with the entries in the index and returns the results. Important: you are **not searching the internet live**, you are searching a

fixed database compiled by someone.

7.2 Seven challenges of online searching

Challenge	Meaning	Example
Spelling variation	British and American spellings differ.	Honour / Honor, Organisation / Organization, Judgement / Judgment, Catalogue / Catalog
Information overload	Too much information; a mix of relevant and non-relevant results; happens when the query is too **general**.	A single general word returning millions of results
Excessive truncation	Cutting the word stem too short retrieves unrelated words.	LIB retrieves Liberty, Libation, Libby, Libel, Library, Librarian
Acronyms	One short form stands for different things; results are based on the letters as words.	ADD (attention deficit disorder), SAD (seasonal affective disorder), AIDS retrieve “add”, “sad”, “aids”
Negation	The database cannot understand “not” in a sentence; it matches the words anyway.	Searching COMPUTER AND OLDER OR ELDERLY returns “we had hoped to include elderly people but...”

Challenge	Meaning	Example
Homographs	Words spelled the same but with different meanings.	Cricket (game) and cricket (insect); KM (kilometre or knowledge management); AGF
Reverse concept	Reversing word order changes the meaning.	MANAGEMENT AND SCHOOL returns “management school” as well as “school management”

⚡ **EXAM HACK**

Seven challenges = **S-I-E-A-N-H-R**: Spelling, Information overload, Excessive truncation, Acronyms, Negation, Homographs, Reverse concept. Cricket = homograph. LIB = truncation. Honour/Honor = spelling. ADD/SAD = acronym. Elderly = negation. Management school = reverse concept.

7.3 Strategies for searching online

- Start with the **closest keywords** and add the place if needed; put the **most important keywords first**.
- Add **synonyms**; in free-text searching use alternative vocabulary.
- Try different search engines.

- If you find a good site, check it for **links to similar sites**.
- Remember: failing to find a term does not mean it does not exist.
- Use **advanced search** features and consider subject-specific alternatives.
- Keep track of where you found useful things.
- Use **more than six words** in a query.
- Use **truncated words** to pick up singular and plural versions.
- Combine keywords into a **phrase** where possible; group concepts with **parentheses**.
- Use **Boolean operators: AND, OR, NOT**.

7.4 Evaluating internet resources

Anyone can publish anything online, and nobody guarantees the credibility of what you find. Check:

- The **title** of the webpage and website.
- The **name and designation** of the author or creator.
- The **purpose and objectives** of the webpage.
- The author's **contact details**.
- The **date produced and last updated**.

Conclusion: search engines retrieve relevant information fast, but they are **not substitutes for**

human expertise.

> ⚡ **EXAM HACK**
> Evaluation = **T-A-P-C-D**: Title, Author, Purpose, Contact, Date. Boolean: AND narrows, OR widens, NOT excludes.

Exam hacks for this chapter (from the CBT)

> ⚡ **EXAM HACK**
> Seven challenges = S-I-E-A-N-H-R: Spelling variation, Information overload, Excessive truncation, Acronyms, Negation, Homographs, Reverse concept. Tie each to its example: honour/honor, too general, LIB, ADD/SAD/AIDS, elderly, cricket/KM, school management vs management school.

> ⚡ **EXAM HACK**
> Boolean: AND narrows (both words), OR widens (either word), NOT removes a word. Truncation too short (LIB) pulls unrelated words.

> ⚡ **EXAM HACK**
> Evaluating a webpage = T-A-P-C-D: Title, Author (name and designation), Purpose, Contact details, Date (produced and last updated).

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. What is online searching?

A. Reading printed books while connected to a library network

B. Retrieving information from online databases by computer

C. Typing a document into a word processor while offline

D. Sending e-mail messages to friends and lecturers

✔ **Answer: B.** It can also mean a specialised search using a string of words entered into a search engine.

2. What does a search engine do when a user types a search request?

A. Sends the request to a librarian to answer by hand

B. Deletes unrelated pages from the internet

C. Compares it with its index and ranks the results

D. Goes live across the whole internet and reads every page

✔ **Answer: C.** The user searches a fixed database compiled by someone, not the live internet.

3. When you use a search engine, what are you actually searching?

A. The shelves of the nearest library
B. A fixed database that has been compiled for the search
C. The hard disk of your own computer
D. The live internet as it is at that moment

✔ **Answer: B.** That is why some new pages do not show up.

4. Searching for 'honour' and missing pages that use 'honor' is an example of which challenge?

A. Homographs
B. Negation
C. Reverse concept
D. Spelling variation

✔ **Answer: D.** British and American spelling: honour/honor, organisation/organization, judgement/judgment, catalogue/catalog.

5. Searching with the truncated stem 'LIB' and getting Liberty, libation and libel is an example of what?

A. Acronyms
B. Information overload
C. Excessive truncation
D. Negation

✔ **Answer: C.** Cutting a word too short pulls in words with different roots.

6. **Which challenge arises when ADD, SAD and AIDS retrieve results based on the words 'add', 'sad' and 'aids'?**

   A. Homographs
   B. Acronyms
   C. Negation
   D. Spelling variation

   ✔ **Answer: B.** The conflation of words to form acronyms confuses the search.

7. **A search for 'computer AND older OR elderly' returns a document saying 'we had hoped to include elderly people'. Which challenge is this?**

   A. Information overload
   B. Negation
   C. Homographs
   D. Reverse concept

   ✔ **Answer: B.** The result contains the words but means the opposite of what was wanted.

8. **The word 'cricket' meaning a game and also an insect is an example of what?**

   A. Homographs
   B. Spelling variation
   C. Negation
   D. Acronyms

   ✔ **Answer: A.** Homographs are words spelt the same but with different meanings, like KM or AGF.

9. Searching 'management AND school' to find 'school management' also returns 'management school'. Which challenge is this?

A. Reverse concept
B. Negation
C. Homographs
D. Acronyms

✔ **Answer: A.** The order of the words changes the meaning.

10. Which of these is a basic strategy for online searching?

A. Always use only one keyword
B. Use the same search engine all the time
C. Never use synonyms of a word
D. Put the most important keywords first

✔ **Answer: D.** Also: start with the closest keywords and mention the place with them.

11. In free-text searching, what should you add to get more results?

A. Synonyms and alternative vocabulary
B. Only words in capital letters
C. Words from another language only
D. Extra spaces between the words

✔ **Answer: A.** Describe the topic with other words too.

12. A search term cannot be found. What should you remember?

A. The information certainly does not exist anywhere
B. You must stop searching online for that topic
C. There may be more; not finding a term proves nothing
D. The search engine is broken and should be replaced

✔ **Answer: C.** Be aware that there is more out there.

13. Which of these are Boolean operators?

A. IF, THEN and ELSE
B. OPEN, CLOSE and SAVE
C. AND, OR and NOT
D. PLUS, MINUS and TIMES

✔ **Answer: C.** Boolean operators combine search words.

14. What is the use of parentheses (brackets) in a search?

A. To hide the search from other users
B. To translate the words to another language
C. To make the results appear faster
D. To distinguish concepts

✔ **Answer: D.** Also combine keywords into a phrase where possible.

15. Which of these is another strategy given for online searching?

A. Forget where you found useful items
B. Avoid subject-specific alternatives
C. Search only with general words
D. Use the advanced search features

✔ **Answer: D.** Also: consider subject-specific alternatives and keep track of useful finds.

16. Why should internet content be evaluated before use?

A. Anyone can publish anything on it
B. All websites are owned by the government
C. Only universities may publish on the internet
D. The internet carries only false information

✔ **Answer: A.** The internet has a lot of valuable information and a lot of junk.

17. Which detail helps you know if a webpage is up to date?

A. Size of the page in kilobytes
B. Name of the browser used
C. Number of visitors to the page
D. Date produced and last updated

✔ **Answer: D.** Dates tell you how current the content is.

18. Why should you check the purpose and objectives of a webpage?

A. To know why the page was created and its use
B. To see who else visited it that day
C. To find out how much money the owner made from it
D. To count the pages on the website

✔ **Answer: A.** Part of evaluating internet resources.

19. Why should you look for the contact details of the author of a webpage?

A. So the page can be printed in colour
B. So you can send him gifts after using his work
C. So the website can load faster
D. So you can reach him to confirm his work

✔ **Answer: D.** It is one of the five tips.

20. A student wants 'school management' and not 'management school'. Which advice helps most?

A. Truncate both words to three letters each
B. Combine the keywords into a phrase
C. Use only the word 'school' in the search box
D. Add as many unrelated keywords as possible

✔ **Answer: B.** Phrase searching keeps the words in the order you want.

CHAPTER 8

Deterioration, Preservation & Conservation

8.1 Why libraries must preserve materials

A library must **acquire, process, organise, preserve and conserve** printed and non-printed materials for present and future users. Resources bought with scarce funds must last. Library information resources include textbooks, journals, magazines, handbooks, directories, maps, gazettes, magnetic films, cassettes, optical discs, video tapes, theses and dissertations.

Signs of deterioration: wear and tear, shrinkage, cracks, **brittleness**, warping, bio-infestation, discolouration, abrasion, holes, dust and dirt.

> **KEY DEFINITION**
>
> **Preservation** — everything that adds to the **physical wellbeing** of library resources, so that protection is included. Broad and preventive.

KEY DEFINITION

Conservation — the **specialised process** of making a fragile object safe, or usable to some extent; operations meant to **prolong the life** of an object by forestalling damage and remedying deterioration.

★ ALZHEIMER'S MODE (the simple version)

Preservation = keeping your clothes clean and in a wardrobe so they never tear. Conservation = taking one torn, fragile wedding dress to a skilled tailor to repair it carefully.

8.2 Three agents of deterioration

Agent	Examples and signs
Environmental	Where the library is built. Swampy area = **reptiles and rodents**. Industrial area = air pollution: heavy **sulphur and nitrogen oxides** make acid that speeds up deterioration. So **survey the land** before building.
Biological (visible)	**Human beings** (mutilation, theft, defacing, mishandling, bad shelving) and **insects**: cockroaches, termites, silverfish, spiders, booklice, bookworms. Signs: white eggs, coloured stains, holes.

Agent	Examples and signs
Microbiological (needs a microscope)	**Fungi, bacteria, moulds**, e.g. *Bacillus* sp. and *Pseudomonas* sp. Cannot be seen by the eye. Regular cleaning prevents dust from settling.

⚡ **EXAM HACK**

Three agents = **E-B-M**. Biological = you can see it (people, insects). Microbiological = you need a microscope (fungi, bacteria, moulds). Cockroach, termite, silverfish = biological. Fungus, mould = microbiological. Swamp = environmental.

8.3 Five inhibitors: why African libraries struggle

Inhibitor	Meaning
Finance	Falling budgets; many libraries cannot even buy resources, let alone preserve them.
Materials	Lack of relevant, working materials and equipment.
Government economic policies	Do not favour libraries; materials are not made locally or are costly due to inflation.
Lack of maintenance culture	Poor upkeep of infrastructure; the reason for quick deterioration.

Inhibitor	Meaning
Lack of technical know-how	Few staff trained in preservation, a specialised field needing knowledge of the physical and chemical nature of materials.

⚡ **EXAM HACK**

Inhibitors = **F-M-G-M-T**: Finance, Materials, Government policies, Maintenance culture, Technical know-how. Money problem comes first.

8.4 Ways to prevent deterioration

- Maintain general **hygiene and cleanliness**.
- **Periodic inspection** of storage areas.
- Be vigilant over new and old acquisitions.
- Use **mechanical traps** instead of toxic chemical bait, so rodents die in the open and not in walls or ceilings.
- Install **air conditioners or fans** to control humidity and keep temperature below **25°C**, which favours microbial growth.
- Discourage users from **wetting fingers with saliva** to turn pages (germs).
- **Appropriate and regular fumigation**.
- **Prohibit edibles** in reading and shelf areas.
- **Train staff** regularly; punish recalcitrant staff or users as a deterrent.

8.5 Principles for restoration

- The restoration manager must have **skills and knowledge** of conservation.
- The **extent of damage** must be known.
- The **numbering** of the document must be checked.
- There must be **records of every document repaired**, safely kept.
- Good **inventory planning and control** of repair materials.

Finally: librarians should conserve resources for the **maximal use of clientele**, which keeps **Ranganathan's Laws of Library Science**.

Exam hacks for this chapter (from the CBT)

> ⚡ **EXAM HACK**
>
> Three agents of deterioration = E-B-M: Environmental, Biological, Microbiological. Biological = visible (humans, insects). Microbiological = needs a microscope (fungi, bacteria, mould).

> ⚡ **EXAM HACK**
>
> Preservation = protect everything that keeps the material healthy (broad, preventive). Conservation = specialised repair to make a fragile item safe and usable (hands-on).

⚡ **EXAM HACK**

Five inhibitors = F-M-G-M-T: Finance, Materials, Government economic policies, Maintenance culture, Technical know-how. Money problem comes first.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **What is the main role of a library with regard to its information resources?**

   A. Lock them away from users to avoid any damage
   B. Give them out permanently to lecturers who ask for them
   C. Acquire, organise and preserve them for users
   D. Sell them to users at a small price to raise funds

   ✔ **Answer: C.** The library must ensure resources bought with scarce funds last for future users too.

2. **Which of these is a sign of deterioration in library materials?**

   A. Colour illustrations
   B. Brittleness
   C. Large print
   D. Fine binding

   ✔ **Answer: B.** Others: wear and tear, shrinkage, cracks, warping, discoloration, abrasion, holes and dirt.

3. Deterioration of library information resources is mainly caused by how many agents?

A. Four
B. Five
C. Three
D. Two

✓ **Answer: C.** Environmental, biological and microbiological.

4. What does polluted air do to library materials?

A. It makes the paper stronger and more flexible
B. It has no effect except on electronic items
C. It removes dust from shelves and books
D. Oxides in it form a damaging acid

✓ **Answer: D.** Libraries near industrial areas suffer from this.

5. Which of these is a biological agent of deterioration?

A. Termites
B. Bacillus species
C. Fungi
D. Bacteria

✓ **Answer: A.** Biological agents: human beings and insects like cockroaches, termites, silverfish, booklice and bookworms.

6. Human beings can damage library materials through which of these?

A. Returning books on time
B. Theft, defacing and mishandling
C. Reading in silence at the reading table
D. Registering with the library

✔ **Answer: B.** Users should be educated on how to handle information resources.

7. Which of these is a microbiological agent of deterioration?

A. Fungi
B. Booklice
C. Silverfish
D. Termites

✔ **Answer: A.** Microbiological agents include fungi, bacteria and moulds.

8. Which of these microorganisms is injurious to library materials?

A. Cockroach
B. Silverfish
C. Bacillus species
D. Termite

✔ **Answer: C.** Bacillus sp. and Pseudomonas sp. are named.

9. Conservation is best described as what?

A. The sharing of books with other libraries
B. The yearly buying of materials for the library
C. Specialised work to make a fragile item usable
D. General cleaning of the library building every morning

✔ **Answer: C.** Operations meant to prolong the life of an object by forestalling damage and remedying deterioration.

10. Which of these is a factor that affects preservation and conservation in African libraries?

A. Large reading rooms
B. Strong teamwork among users
C. Finance
D. Free supply of equipment

✔ **Answer: C.** Budgets for most African libraries are going down.

11. Why do government economic policies affect preservation in Africa?

A. The government supplies all equipment free of charge
B. Materials are not made locally or cost too much
C. Libraries are banned from buying anything abroad
D. Libraries are made to pay no tax on materials

✔ **Answer: B.** Policies in most African countries do not favour libraries.

12. What does a lack of maintenance culture lead to?

A. Quick deterioration of the collections
B. Better use of the library by students
C. Lower cost of buying new books
D. Longer life for the books on the shelves

✔ **Answer: A.** Poor maintenance of infrastructure is a reason for quick deterioration.

13. Why is technical know-how important for preservation?

A. It reduces the number of users in the library
B. It removes the need for any equipment
C. It is a specialised field needing skills
D. It helps staff to catalogue books faster

✔ **Answer: C.** Staff must have relevant skills for a successful programme.

14. Which trap is preferred for rodents in a library?

A. Poisoned food on the shelves
B. Mechanical traps
C. Toxic chemical bait
D. Strong perfume sprays

✔ **Answer: B.** Mechanical traps are preferred so that the rodents die in the open, not inside walls or ceilings.

15. Why are air conditioners or fans installed in storage areas?

A. To keep the reading tables cold for students
B. To dry the ink on freshly printed books
C. To make the room smell better for visitors
D. To control humidity and temperature

✔ **Answer: D.** Gives 25 degrees Celsius as favourable to microbial growth.

16. What should be done regularly to control insects in the library?

A. Moving the books outdoors in the sun
B. Appropriate fumigation
C. Painting the walls with oil paint
D. Spraying water on the shelves

✔ **Answer: B.** Fumigation is done on a regular, appropriate schedule.

17. Taking edibles into reading or shelf areas is what?

A. Allowed if users clean up after themselves
B. Allowed during examination periods
C. Prohibited, since it attracts pests
D. Allowed for staff only, at any time

✔ **Answer: C.** Library hygiene also includes training staff and punishing recalcitrant users.

18. Which of these is a principle for restoring information resources?

A. The extent of the damage need not be known
B. The numbering of loose documents need not be checked
C. The manager must know conservation and restoration
D. No records of repairs should be kept

✔ **Answer: C.** The extent of damage must be known and records of repairs safely kept.

19. Reptiles and rodents are most likely to find their way into a library built where?

A. Beside a well-kept garden
B. On dry and elevated ground
C. In a swampy area
D. Inside a busy school compound

✔ **Answer: C.** The environment where a library is sited affects the life span of its materials.

20. What should library staff receive regularly to help preservation and conservation?

A. Training on conservation and hygiene
B. Lessons on how to sell damaged books
C. Extra holidays after cleaning the shelves
D. Gifts for handling very old books carefully

✔ **Answer: A.** Recalcitrant staff or users can be punished as a deterrent.

CHAPTER 9

ICT: Threats, Networks, Internet, IoT & NFTs

9.1 Computer threats

★ ALZHEIMER'S MODE (the simple version)

A threat is anything that can harm your computer or data. It can be **intentional** (a hacker stealing bank details) or **accidental** (a computer soaked by rain, a fault). Malicious code is a harmful program, also called a **rogue program**.

KEY DEFINITION

Malicious code — a threat to computing assets caused by a programmer's code written to damage. The agent is the **writer of the code** or anyone who distributes it.

Code	What it does	Remember
Virus	Passes malicious code to other programs by **modifying them**. Attaches to files like **.doc, .xls, .exe**. Can take over the system and spread.	Needs a host file

Code	What it does	Remember
Trojan horse	Looks useful but does unexpected things. Must be **installed** by a user or intruder first. Example: a fake login script that steals your ID and password.	Pretends to be helpful
Logic bomb	Goes off when a **specific condition** occurs. Example: **time bomb** (date or time).	Waits for a trigger
Trapdoor / Backdoor	A **secret, undocumented** way into a program with special privileges. Can be benign (troubleshooting) or used for illegal access.	Secret entrance
Worm	**Standalone** program that copies itself and spreads through a **network**, using security failures. Always uses up **bandwidth**.	Travels alone

Worm	Virus
Operates through the **network**	Spreads through **any medium**
Spreads copies as a **standalone program**	Spreads copies as a program that **attaches to other programs**
Almost always harms the network (uses bandwidth)	Almost always corrupts or modifies files on the target computer

> ⚡ **EXAM HACK**
> Worm = **W**ire/network, alone. Virus = **V**ictim files, needs a host. Trojan = **fake friend**. Logic/time bomb = **waits for a date or condition**. Backdoor = **secret door**.

Security measures

Tool	Job	Examples
Antivirus	Identifies and removes viruses in memory, on storage media and in incoming e-mail. Watches for programs trying to modify the boot program or operating system.	Norton, ESET, Kaspersky
Anti-spy ware	Removes **spyware**, a program placed without your knowledge that secretly collects information about you. Also called tracking software or a spybot.	Spybot Search and Destroy, Ad-aware, Spyware Blaster
Firewall	Hardware or software in a network that **blocks communications forbidden** by the security policy; may permit limited access.	—

Data protection: **make backup files**, **detect viruses and clean the computer**, and **warn others** about virus attacks. **Backup** is copying data from secondary storage into an archive so it can be restored after a loss. It is a simple kind of disaster recovery but **not a**

complete plan on its own.

> ⚡ **EXAM HACK**
> Antivirus names: Norton, ESET, Kaspersky. Anti-spyware names: Spybot, Ad-aware, Spyware Blaster. They are not the same list. Data protection = **B-D-W**: Backup, Detect and clean, Warn.

Operating systems: Microsoft Windows (XP, 2000, 7, 8, 8.1, 10), Linux, Ubuntu, Fedora, Macintosh (Mac).
Office packages: Microsoft Office (Word, PowerPoint, Excel, Access) and LibreOffice (on Ubuntu).

9.2 What is ICT?

> KEY DEFINITION
> **ICT (Information and Communication Technology)** — technology for information processing: electronic computers, communication devices and software used to convert, store, protect, process, transmit and retrieve information from anywhere, anytime. It also covers telecommunications, broadcast media, intelligent building management, audiovisual systems and network-based control.

ICT is often called an extended synonym of IT, but its scope is **broader** because it merges telephone and audiovisual networks with computer networks.

9.3 Computer networks

KEY DEFINITION

Computer networking — the techniques, physical connections and programs used to **link two or more computers** so they can share resources (hard discs, modems, printers).

- Two types of connection: **physical** (cables and signals; defined by medium, topology and method) and **logical/virtual** (protocols that let apps like e-mail and browsers exchange information).
- Three basic components: **network software, application software, network hardware**.
- Networking devices: network interface cards, hubs, switches, routers, modems, uninterruptible power supplies, wiring and cable.

Cable	Note
Twisted-pair	Industry standard in new installations (UTP = unshielded twisted pair)
Coaxial	Looks like round cable-TV wiring
Fibre-optic	Mostly for **backbone** devices; the **most reliable and most expensive**

Network	Coverage
LAN (Local Area Network)	One building or one campus, within a few kilometres. Speeds are much **higher than WAN**.

Network	Coverage
WAN (Wide Area Network)	Long distances via telephone lines, fibre optic or satellite. The **Internet is a super WAN**.
MAN (Metropolitan Area Network)	A city-sized network.
CAN (Campus Area Network)	An educational or corporate campus; larger than a LAN; usually several LANs joined by switches and routers. At FUOYE it is **fuoyeNET** (only campus residents).
PAN (Personal Area Network)	Around one person, e.g. phone or watch with a laptop (Bluetooth).

Topology (how computers are connected): **Bus, Ring, Star**.

Wireless vs wired

	Wired	Wireless
Reliability and security	**Most reliable and most secure**	Prone to interference; good if encrypted
Mobility	Limited	**Outstanding**
Installation	Moderate difficulty	Easy, but beware interference
Cost	Less	More

- Wireless networks use **radio waves**. Examples: **Wi-Fi (Wireless Fidelity)** and **Bluetooth**.
- **SSID** = Service Set IDentifier = the **name of a Wi-Fi network**. FUOYE's is **fuoyenet**.

- **WPA / WPA2** = Wi-Fi Protected Access: a wireless security standard. The WPA2 key is the **password** to your connection.

9.4 The Internet and the Web

- The **Internet** is a global network that uses **TCP/IP**; it is the world's largest network.
- The **Web (WWW)** is **one service** on the internet. Internet = infrastructure. Web = a service. They are not the same.
- Roots: **ARPANET** (1960s, US government) → **NSFNET** (1985, National Science Foundation) → commercial providers (1995).
- **IP** addresses (IPv4, IPv6) identify devices. Services: e-mail, video conferencing, **FTP** file sharing, instant messaging, forums, social networks, online shopping.
- **Browsers** locate and display web information; **web portals** act as a home base and gateway.

9.5 Internet of Things (IoT)

★ ALZHEIMER'S MODE (the simple version)

IoT = everyday **things** (a car, a door lock, a watch, even a farm animal) with sensors that can **send data over the internet on their own**, without a person pressing buttons.

KEY DEFINITION

Internet of Things — a system of interrelated devices, machines, objects, animals or people with **unique identifiers (UIDs)** that can transfer data over a network without human-to-human or human-to-computer interaction.

- How it works: smart devices with sensors collect data, send it to an **IoT gateway**, then data is analysed in the cloud or locally; devices act on it.
- Benefits: monitor business processes, better customer experience, save time and money, raise productivity, better decisions, more revenue.
- Most common in manufacturing, transportation and utilities; also agriculture, infrastructure, home automation, healthcare.

IoT in education	What it does
Evolving met hodologies	E-books, smart boards, central server controlling syllabus, voice command, speech-to-text notes, GPS-tracked school buses, disaster alarms.
Automated attendance	Biometric or barcode ID cards record attendance automatically; parents can be messaged about absence; staff entry and exit times tracked.

IoT in education	What it does
Safety in premises	Cameras, fire and short-circuit sensors with exact location, smart door locks and automatic alerts.
Distance learning	Live and recorded classes, timed online assessments, tracking time spent on a portal.

Example	What it is
Scanmarker	Scans text from books into a phone or computer and translates it into 40+ languages.
Blackboard	A connected learning experience and support network.
Magicard	Student smart cards: access, photo ID, attendance, payment, health data.
Kajeet	Wi-Fi systems for **smart school buses**; parents track the bus in real time.
LocoRobo	LocoIoT course teaching students to build and connect IoT systems.

9.6 E-commerce, blockchain, cryptocurrency and NFTs

Term	Meaning
E-commerce	Buying and selling goods using the internet. Customers, suppliers and employees all benefit from ICT.

Term	Meaning
Blockchain	A **decentralised ledger** of all transactions across a **peer-to-peer network**; participants confirm transactions **without a central clearing authority**.
Cryptocurrency	Digital medium of exchange that uses **encryption** to control creation of units and verify transfers. **Bitcoin** is the best known (blockchain was invented for it). Others: Ethereum, Tether (USDT), Litecoin, XRP, Binance Coin, Cardano, Polkadot.
NFT (Non-Fungible Token)	A programmable, **unique** unit of data that **cannot be changed** once recorded on a blockchain. Created by **minting** (pay-per-use). Marketplaces: **OpenSea** (first and largest), Mintable, Nifty Gateway, Rarible.

- Uses of NFTs: digital art, digital music, virtual real estate, game objects, video clips, upselling physical products like sneakers.
- Buying an NFT means you own an **encrypted token** on a blockchain that represents the object. You do **not** automatically own the copyright or IP.
- Artwork vs NFT: artwork can be altered by its owner and has watermarks; an NFT **cannot be altered** and has unique encrypted code. Both get value from supply and demand.

⚡ **EXAM HACK**

NFT = unique, unalterable, one-of-a-kind. Blockchain = the shared ledger. Cryptocurrency = digital money (Bitcoin first). Minting = creating an NFT. Buying an NFT is not buying copyright. IoT = **UID + data over a network with no human step**.

Exam hacks for this chapter (from the CBT)

⚡ **EXAM HACK**

Malicious code family: Virus (attaches to files, needs a host), Worm (stands alone, travels through the network), Trojan horse (looks useful, tricks you into installing), Logic bomb (waits for a condition or date), Trapdoor/Backdoor (secret entry). Worm = Network, Virus = Files.

⚡ **EXAM HACK**

Three network sizes to remember: LAN (one building or campus), CAN (whole campus, e.g. fuoyeNET), WAN (long distance, the Internet is a super WAN). Topologies are Bus, Ring, Star.

⚡ **EXAM HACK**

NFT = unique, cannot be altered, one-of-a-kind (Non-Fungible). Cryptocurrency = digital money. Blockchain = the shared ledger that records both. Buying an NFT does not mean you own the copyright.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **A computer threat that comes from hacking is described as what?**

   A. Intentional
   B. Accidental
   C. Harmless
   D. Natural

   ✔ **Answer: A.** Intentional (hacking) or accidental (malfunction or physical damage).

2. **Malicious code is also known as what?**

   A. A search engine
   B. A backup file
   C. A firewall
   D. A rogue program

   ✔ **Answer: D.** It threatens computing assets through damage caused by a programmer's code.

3. What is a Trojan horse?

A. A program that looks useful but does unexpected things once installed
B. A program that destroys files when a certain date arrives
C. A program that blocks unwanted network traffic
D. A program that copies itself across a network without a host file
✔ **Answer: A.** A fake login script that steals your ID and password is an example.

4. A time bomb is an example of what?

A. Worm
B. Trojan horse
C. Logic bomb
D. Firewall
✔ **Answer: C.** A logic bomb goes off when a specific condition or time is reached.

5. What is the main difference between a worm and a virus?

A. A worm attaches to other programs, while a virus spreads over the network alone
B. A worm attacks only hardware, while a virus attacks only software
C. A worm is harmless, while a virus always destroys the screen
D. A worm spreads over the network alone; a virus attaches to programs
✔ **Answer: D.** Worms consume bandwidth; viruses corrupt or modify files.

6. Which of these is an antivirus program?

A. SpywareBlaster
B. Kaspersky
C. Ad-aware
D. Spybot

✔ **Answer: B.** Norton, ESET and Kaspersky are antivirus. Spybot, Ad-aware and SpywareBlaster are anti-spyware.

7. What does a firewall do?

A. Copies computer data into an archive file
B. Removes viruses from stored e-mail files
C. Blocks forbidden communications in a network
D. Records everything a user types on the keyboard

✔ **Answer: C.** It can be hardware or software and may permit limited access.

8. What is a data backup?

A. Sending a copy of a virus to other users
B. Deleting old files to free space on the hard drive
C. Switching off the computer after long use
D. Copying data into an archive for restoring later

✔ **Answer: D.** Backups are a simple kind of disaster recovery but not a complete plan on their own.

9. What is computer networking?

A. Sending e-mails to a computer at home only
B. Linking computers to share resources
C. Printing documents from one computer to another
D. Storing programs in the memory of one computer

✔ **Answer: B.** Shared resources include hard discs, printers and modems.

10. Which of these is a basic component of a classical computer network?

A. Network hardware
B. A telephone directory
C. A paper catalogue
D. A card index

✓ **Answer: A.** The three are network software, application software and network hardware.

11. Which type of network spans long distances using telephone lines, fibre-optic cables or satellite links?

A. Local Area Network (LAN)
B. Wide Area Network (WAN)
C. Campus Area Network (CAN)
D. Personal Area Network (PAN)

✓ **Answer: B.** The Internet is a super WAN. LAN speeds are higher than WAN speeds.

12. What is a Campus Area Network (CAN)?

A. A network covering only one room or floor
B. A network linking phones and watches to a laptop
C. A network covering a school or business campus
D. A network linking computers across countries

✓ **Answer: C.** It is usually made of several LANs connected by switches and routers.

13. Which statement about wired and wireless networks is correct?

A. Wired and wireless are the same in cost and reliability
B. Wireless needs cables but wired needs none
C. Wireless is the most reliable and secure, while wired gives outstanding mobility
D. Wired is the most reliable and secure, while wireless gives outstanding mobility
✔ **Answer: D.** Wireless is more expensive, easier to install, but open to interference.

14. What is an SSID?

A. The address of a website
B. The speed of a Wi-Fi network
C. The name of a Wi-Fi network
D. The password of a Wi-Fi network
✔ **Answer: C.** Service Set IDentifier. FUOYE's is fuoyenet.

15. How is the Internet different from the World Wide Web?

A. The internet is the global communication system, while the web is one service on it
B. The internet is only for e-mail and the web only for games
C. They are the same thing and cannot be told apart
D. The web is the global communication system, while the internet is one service on it
✔ **Answer: A.** The internet uses TCP/IP to transmit data.

16. Which network did the U.S. government start building in the 1960s that led to the internet?

A. FUOYENET
B. NSFNET
C. BITNET
D. ARPANET

✔ **Answer: D.** NSFNET came in 1985, and commercial providers took over around 1995.

17. How can IoT help with student attendance?

A. Lecturers call the roll by hand at the end of class
B. Students write their names in a register at the gate
C. Students sign in using paper slips
D. Attendance is taken using ID cards and biometrics

✔ **Answer: D.** Staff entry and leaving times can also be tracked.

18. Kajeet makes Wi-Fi systems for what?

A. Smart school buses
B. Smart classrooms
C. Library shelves
D. Student hostels

✔ **Answer: A.** Students use the bus Wi-Fi to finish homework, and parents can track the bus in real time.

19. What is a blockchain?

A. A central bank register controlled by one clearing authority

B. A paper record of transactions kept in a vault

C. A decentralised ledger on a peer-to-peer network

D. A computer program for sorting e-mails

✔ **Answer: C.** It lets participants confirm transactions without a central clearing authority.

20. Bitcoin is best known as what?

A. The best-known cryptocurrency

B. A bank account used by the government

C. A type of firewall for networks

D. A program for cleaning viruses

✔ **Answer: A.** Others: Ethereum, Tether, Litecoin, XRP, Binance Coin, Cardano.

CHAPTER 10

Computers: Classification, Components & Supercomputers

10.1 What is a computer?

KEY DEFINITION

Computer — an electronic device that accepts **data** as input (keyboard, mouse, joystick, camera), **processes** it (CPU), **stores** it if needed, and gives out **output** (monitor, printer, speaker) known as **information**, according to instructions called a **program**.

- **GIGO = Garbage In, Garbage Out**: the quality of the output cannot be better than the quality of the input.
- Three major operations: **IPO** = Input, Process, Output. With communication and storage added = **IPOCS**.

10.2 Computer history: the timeline

★ ALZHEIMER'S MODE (the simple version)

Computers did not appear suddenly. People first counted with beads, then with gears, then with punched cards, then with electricity. Learn the line in order and you win.

Year	Device and inventor	What it did
~500 BC	**Abacus**	First counting machine: wires, frames, beads. Addition and subtraction.
1617	**Napier's Bones** (John Napier, Scottish)	Multiplication and division using addition and subtraction.
1632	**Slide rule** (England)	Analog computer for roots, logarithms and trigonometry; used until the pocket calculator.
1642	**Pascaline** (Blaise Pascal, age 19)	Gear-driven; helped his tax-collector father; could **only add**.
11 years after Pascal's death	**Leibniz multiplier / Stepped Reckoner** (German)	Add, subtract, multiply, divide, square roots.
—	**Jacquard's loom**	**Punched cards** controlled weaving patterns.

Year	Device and inventor	What it did
1822	**Difference Engine** (Charles Babbage, British)	Calculate mathematical values automatically; about 25,000 parts, 15 tons; never completed.
1833	**Analytical Engine** (Babbage)	Punched cards for input, memory unit, a “mill” like a CPU, conditional statements. Babbage = **father of the computer**.
1890	**Hollerith census machine**	Census tabulating; company became **IBM**.

20th century onwards: ENIAC = Electronic Numerical Integrator and Calculator (first general-purpose electronic computer; vacuum tubes; 1000 times faster than earlier machines). **EDVAC** = Electronic Discrete Variable Automatic Computer (single processing unit; architect of modern computers). **UNIVAC** = Universal Automatic Computer (magnetic tape for non-volatile storage). All three are **first generation**.

Personal computers by form

- **Desktop:** system unit (casing with CPU), monitor, keyboard, mouse.
- **Laptop:** compact; runs on AC and battery; keyboard, LCD screen, built-in pointing device.

- **Notebook:** very small, usually under **6 lb (2 kg)**; fits in a briefcase; smaller than a laptop.
- **Palmtop:** hand-held, no keyboard; the screen is the input; uses a **stylus pen**; fits in a pocket.

> ⚡ **EXAM HACK**
>
> Timeline: **Abacus (500 BC) → Napier (1617) → Slide rule (1632) → Pascaline (1642) → Leibniz → Jacquard → Babbage (1822, 1833) → Hollerith (1890) → ENIAC**. Pascal = add only. Leibniz = all four operations. Babbage = father. Hollerith = IBM.

10.3 Four ways to classify computers

Computers are classified by **Generation, Types, Size and Purpose** (G-T-S-P).

By	Groups	Details
Type	Analog, Digital, Hybrid	**Analog**: continuous data (voltmeter, slide rule, temperature gauges; fuel pump control). **Digital**: binary 0 and 1 (desktop, laptop, palmtop). **Hybrid**: both (mini, mainframe, supercomputers).

By	Groups	Details
Size	Super, Mainframe, Mini, Micro	**Super**: fastest; weather forecasting, designing cars, aircraft, spacecraft. **Mainframe**: many programs at once, huge data in petabytes, ATM, replaces dozens of servers. **Mini**: medium size, many terminals, shared; supermarkets, airlines, cinemas. **Micro**: desktop, tower, laptop, notebook, palmtop.
Purpose	Special, General	**Special**: built for one application (traffic control, weapon guidance). **General**: many applications in homes, schools, offices.

10.4 Components of a computer system

- **Hardware**: physical parts you can see and touch: **input, storage, processing, output**.
- **Input** (keyboard, joystick, stylus, mouse) → **Storage** (primary and secondary) → **Processing** by the **CPU** = **Control Unit (CU) + Arithmetic and Logic Unit (ALU)** → **Output** (monitor = soft copy; printer = hard copy).
- **Software**: the instructions that direct the hardware.

10.5 Fastest supercomputers (top 10 list)

#	Name	Speed	Country / maker
1	**Fugaku**	442 Pflop/s	Japan, Fujitsu, RIKEN (Kobe)
2	**Summit**	148.8 Pflop/s	USA, IBM, Oak Ridge (fastest in the US)
3	**Sierra**	94.6 Pflop/s	USA, Lawrence Livermore
4	**Sunway TaihuLight**	93 Pflop/s	China, Wuxi
5	**Perlmutter**	64.6 Pflop/s	USA, new entry
6	**Selene**	63.4 Pflop/s	USA, Nvidia
7	**Tianhe-2A**	61.4 Pflop/s	China, NUDT, Guangzhou
8	**JUWELS Booster**	44.1 Pflop/s	Germany, Atos; most powerful in Europe
9	**HPCS**	35.5 Pflop/s	Italy, Dell, Eni S.p.A.
10	**Frontera**	23.5 Pflop/s	USA, Dell, Texas

⚡ **EXAM HACK**

Top 5 = **Fugaku (Japan), Summit (USA), Sierra (USA), Sunway TaihuLight (China), Perlmutter (USA)** = J-U-U-C-U. Summit = fastest in the US. Some notes call Summit “the fastest” (it is the fastest in the US), but the ranked list puts Fugaku first. Fugaku wins.

Exam hacks for this chapter (from the CBT)

⚡ **EXAM HACK**

Timeline in order: Abacus (500 BC) > Napier's Bones (1617) > Slide Rule (1632) > Pascaline (1642) > Leibniz > Jacquard loom > Babbage Difference Engine (1822) > Analytical Engine (1833) > Hollerith (1890) > ENIAC.

⚡ **EXAM HACK**

Four ways to classify a computer = G-T-S-P: Generation, Types (analog, digital, hybrid), Size (super, mainframe, mini, micro), Purpose (special, general).

⚡ **EXAM HACK**

Top 5 supercomputers in order of speed: Fugaku (Japan) > Summit (USA) > Sierra (USA) > Sunway TaihuLight (China) > Perlmutter. Japan, USA, USA, China, USA.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **Which of these is the best description of a computer?**

   A. A device that stores programs but cannot process data
   B. A device that takes in, processes and gives out data
   C. A telephone network that links many offices
   D. A machine that only prints documents on paper
   ✔ **Answer: B.** Input, process, output and storage, following instructions called a program.

2. **What does GIGO mean?**

   A. Garbage In, Garbage Out
   B. Good Input, Good Output
   C. General Input, General Output
   D. Graphics In, Graphics Out
   ✔ **Answer: A.** The quality of the output cannot be better than the quality of the input.

3. **What are the three major operations of a computer called IPO?**

   A. Input, Process and Output
   B. Input, Print and Open
   C. Instruction, Process and Order
   D. Internet, Program and Output
   ✔ **Answer: A.** IPOCS adds Communication and Storage.

4. John Napier's bones were invented to do what?

A. Count census figures for the government
B. Print mathematical tables automatically
C. Multiply and divide by simple addition
D. Weave patterns on fabric using punched cards

✓ **Answer: C.** Napier noticed this in 1617.

5. Which device was first built in England in 1632 and used for roots, logarithms and trigonometry?

A. Slide rule
B. Abacus
C. Difference Engine
D. Pascaline

✓ **Answer: A.** It is an analog computer that engineers used until the pocket calculator came.

6. Which machine was called the 'Stepped Reckoner'?

A. Leibniz multiplier
B. Hollerith machine
C. Pascaline
D. Napier's bones

✓ **Answer: A.** It could add, subtract, multiply, divide and find square roots.

7. What did Jacquard's loom contribute to computing?

A. Gears that added up numbers
B. Vacuum tubes that stored data
C. Punched cards to guide weaving
D. Beads used to count numbers

✔ **Answer: C.** Descendants of the punched cards have been used ever since.

8. Why is Charles Babbage called the father of the computer?

A. He built the first microcomputer with a keyboard and screen
B. He invented the abacus and the slide rule together
C. His Analytical Engine had a design close to modern computers
D. He started the company that later became IBM

✔ **Answer: C.** The Analytical Engine had a memory unit, a mill like the CPU and a conditional statement.

9. Herman Hollerith's census machine led to the creation of which company?

A. Apple
B. Fujitsu
C. Microsoft
D. IBM

✔ **Answer: D.** It was used for the 1890 census and later grew into International Business Machines.

10. What does EDVAC stand for?

A. Electric Data Value and Control
B. Electronic Digital Visual Analysis Computer
C. Electronic Discrete Variable Automatic Computer
D. Electronic Direct Variable Access Computer

✔ **Answer: C.** It used a single processing unit and is called the architect of modern computers.

11. UNIVAC used what for non-volatile storage?

A. Punched cards
B. Flash drive
C. Magnetic tape
D. Floppy disk

✔ **Answer: C.** It was a first generation computer and stands for Universal Automatic Computer.

12. A notebook computer usually weighs less than what?

A. 6 pounds (about 2 kg)
B. 1 pound (about 0.5 kg)
C. 20 pounds (about 9 kg)
D. 10 pounds (about 4.5 kg)

✔ **Answer: A.** It fits inside a briefcase and is smaller than a laptop.

13. Which type of computer processes data in a continuous form?

A. Digital
B. Analog
C. Mainframe
D. Hybrid

✔ **Answer: B.** Examples: voltmeter, slide rule and temperature gauges.

14. Which type of computer works with data in binary code, using 0 and 1?

A. Hybrid
B. Analog
C. Digital
D. Abacus

✔ **Answer: C.** Examples: desktop, laptop and palmtop.

15. Supercomputers are mainly used to do what?

A. Store personal photos for students
B. Play music and videos in homes
C. Solve complex problems like weather forecasts
D. Type letters and print documents in offices

✔ **Answer: C.** They also design cars, aircraft and spacecraft.

16. Mainframe computers are known for doing what?

A. Fitting inside a pocket for personal use
B. Running only one program at a time
C. Running many programs at the same time
D. Working only as a calculator

✓ **Answer: C.** A single mainframe can replace dozens or hundreds of smaller servers.

17. A computer built for only one application, like traffic control, is called what?

A. General purpose computer
B. Analog computer
C. Special purpose computer
D. Hybrid computer

✓ **Answer: C.** Weapon guidance systems are another example.

18. Computers used in homes, schools and offices for many different applications are called what?

A. Special purpose computers
B. General purpose computers
C. Supercomputers
D. Analog computers

✓ **Answer: B.** They can be multi-media or internet-ready.

19. Which two parts make up the CPU?

A. Control Unit and Arithmetic Logic Unit
B. Hard disk and flash drive units
C. Monitor, speaker and printer units
D. Keyboard, mouse and scanner units

✔ **Answer: A.** The CPU is the unit through which all processing is carried out.

20. Which was the fastest supercomputer in the US in the list?

A. Tianhe-2A
B. Sunway TaihuLight
C. Fugaku
D. Summit

✔ **Answer: D.** Summit is built by IBM and sits at Oak Ridge National Laboratory in Tennessee.

FUOYE SECTION

FUOYE Website, Portal, Email & Campus Network

11.1 Website, portal and email

★ ALZHEIMER'S MODE (the simple version)

Three doors to FUOYE online: the **website** (information about the school), the **portal** (where YOU do your registration, courses and payments), and your **school email** (how the school contacts you). Know all three addresses.

Item	Detail
Main website	**https://fuoye.edu.ng**
University portal	The site where the main student activities are done: **registration, courses and payment**. **https://ecampus.fuoye.edu.ng** **https://portal.fuoye.edu.ng** Each student must visit these sites **regularly** for information.
Student email format	**firstname.lastname.matricno@fuoye.edu.ng** or **lastname.firstname.matricno@fuoye.edu.ng** (Example: charles.ojo.12345@fuoye.edu.ng)

Item	Detail
Default email password	**fuoye123**. You must **change it**.
Wi-Fi name (SSID)	**fuoyenet**
Campus network (CAN)	**fuoyeNET**: only campus residents (students and faculty) can access it. Also called a residential network or ResNet.
Library software	**Koha** (open source) is used at FUOYE.
Library address	P.M.B. 373, Oye-Ekiti, Ekiti State, Nigeria. Ownership stamp: Federal University Oye-Ekiti Library.

⚡ **EXAM HACK**

Website = fuoye.edu.ng. **Portal** = ecampus. and portal. fuoye.edu.ng. **Email** ends @fuoye.edu.ng and holds your **name + matric number** in a set order. Password starts as **fuoye123** and must be changed. **Wi-Fi** = fuoyenet.

⚠ **TRAP ALERT**

Many wrong options change one small thing: .com instead of .edu.ng, www.ecampus.fuoye.ng, or .org.ng. The correct domain always ends in **.edu.ng**.

Exam hacks for this section

⚡ **EXAM HACK**

Portal sites (registration, courses, payments) = ecampus.fuoye.edu.ng and portal.fuoye.edu.ng. Main website = fuoye.edu.ng. Email ends @fuoye.edu.ng. Wi-Fi name = fuoyenet.

⚡ **EXAM HACK**

Default email password is fuoye123, and you MUST change it. Login format has your name and matric number inside, in a set order.

⚡ **EXAM HACK**

FUOYE library runs Koha (open source). The campus network (CAN) only residents can use is called fuoyeNET.

Trap questions (20)

Try each one first. Cover the answer line with your finger, choose A, B, C or D, then check. The reason under every answer is the part to remember.

1. **What is the main website address of the Federal University Oye-Ekiti?**

   A. https://ecampus.fuoye.edu.ng
   B. https://library.fuoye.edu.ng
   C. https://fuoye.edu.ng
   D. https://portal.fuoye.edu.ng

   ✔ **Answer: C.** Gives the main site as fuoye.edu.ng.

2. What is the university portal used for?

A. Reading newspapers and journals online
B. Registrations, courses and payments
C. Borrowing books from the library
D. Watching lectures that were recorded in class

✔ **Answer: B.** It is the site where the main activities of students are done.

3. How often is each student expected to visit the portal sites?

A. Only during registration
B. Regularly, for information
C. Once in every session
D. Only after results are released

✔ **Answer: B.** Says each student is expected to visit these sites for information regularly.

4. A student's FUOYE email address ends with which domain?

A. @fuoye.com
B. @fuoye.edu.ng
C. @fuoye.org.ng
D. @gmail.fuoye.ng

✔ **Answer: B.** Example : charles.ojo.12345@fuoye.edu.ng.

5. What is the default password for a new FUOYE student email?

A. password
B. oyeekiti1
C. fuoye2024
D. fuoye123

✔ **Answer: D.** The default password must be changed after first login.

6. What must a student do with the default email password?

A. Change it
B. Keep it unchanged
C. Write it on the ID card
D. Share it with the department

✔ **Answer: A.** Says the default password has to be changed.

7. What does a campus area network in FUOYE go by?

A. fuoyeNET
B. oyeNET
C. ecampusNET
D. portalNET

✔ **Answer: A.** It can only be accessed by campus residents such as students and faculty.

8. Who can access fuoyeNET?

A. Only visitors with a guest pass
B. Only the staff of the ICT Directorate
C. Only campus residents
D. Anyone in the world with a phone

✔ **Answer: C.** It works like a 'residential network' or 'ResNet'.

9. Which open source library software is used in the FUOYE Library?

A. Aleph
B. Koha
C. Symphony
D. Millennium

✔ **Answer: B.** Koha is the open source software named for FUOYE.

10. Which stamp is placed on materials in the university library as the institution's ownership stamp?

A. Library of Congress
B. Ministry of Education Library
C. Federal University Oye-Ekiti Library
D. National Universities Commission

✔ **Answer: C.** Stamping is the first step of processing after acquisition and verification.

11. FUOYE Library is an example of which type of library?

A. Public library
B. School library
C. National library
D. Academic library

✔ **Answer: D.** Academic libraries are attached to tertiary institutions like FUOYE and EKSU.

12. In what state is the Federal University Oye-Ekiti located?

A. Kwara State
B. Ondo State
C. Ekiti State
D. Osun State

✔ **Answer: C.** Addresses use Oye-Ekiti, Ekiti State.

13. Which of these is a FUOYE portal address given?

A. https://results.fuoye.edu.ng
B. https://news.fuoye.edu.ng
C. https://ecampus.fuoye.edu.ng
D. https://studentlife.fuoye.edu.ng

✔ **Answer: C.** The two portal sites are ecampus.fuoye.edu.ng and portal.fuoye.edu.ng.

14. Which of these is also a FUOYE portal address given?

A. https://portal.fuoye.edu.ng
B. https://fees.fuoye.edu.ng
C. https://hostel.fuoye.edu.ng
D. https://exams.fuoye.edu.ng

✔ **Answer: A.** Each student is expected to visit these sites regularly for information.

15. Which details make up a FUOYE student email login example?

A. First name, last name and matric number
B. Department, level and phone number
C. Surname, state of origin and date of birth
D. Faculty, hostel name and surname only

✔ **Answer: A.** Format: firstname.lastname.matricno@fuoye.edu.ng.

16. says the email login can also be written in what other order?

A. Matric number, department and first name
B. Matric number, first name and level
C. Last name, first name and matric number
D. Hostel name, last name and first name

✔ **Answer: C.** The alternative is lastname.firstname.matricno@fuoye.edu.ng.

17. What is the SSID of the school Wi-Fi given?

A. fuoyecampus
B. fuoyenet
C. fuoyewifi
D. oyeekitinet

✔ **Answer: B.** An SSID is the name of a Wi-Fi network.

18. What is the P.M.B. number given for the Federal University Oye-Ekiti library?

A. P.M.B. 373
B. P.M.B. 137
C. P.M.B. 733
D. P.M.B. 273

✔ **Answer: A.** The address is P.M.B. 373, Oye-Ekiti, Ekiti State, Nigeria.

19. Which unit of FUOYE wrote chapters on ICT and networks in the library course?

A. The Security Unit and the Works Department
B. ICT Directorate and Science Education
C. The Student Affairs and Alumni Office
D. The Bursary and the Registry

✔ **Answer: B.** One chapter was written by the ICT Directorate, another by Science Education (Educational Technology Unit).

20. sample email charles.ojo.12345@fuoye.edu.ng, which part is the student's last name?

A. 12345
B. ojo
C. charles
D. fuoye

✔ **Answer: B.** Format is firstname.lastname.matricno, so charles is the first name and 12345 is the matric number.

PAST QUESTIONS

Marked Past-Question Bank (60)

Every question below comes from the past-question file, with the answer exactly as marked in it. Cover the answer, attempt, then check. The same questions are in the CBT under **Past Questions**.

Past questions with marked answers

⚡ EXAM HACK

'All of these EXCEPT' questions: tick every option that is true, the one left over is your answer. Do not stop at the first true option you see.

⚡ EXAM HACK

Full forms: ISSN = Serials (S for Serial), ISBN = Book (B for Book), OPAC = Online Public Access Catalogue, SSID = Service Set IDentifier, LAN = Local, WAN = Wide.

⚡ EXAM HACK

Call mark = class number + cutter number, pasted on the spine. Accession number = unique serial number given on arrival. Do not swap the two.

1. **What is the FUOYE website?**

   A. www.ecampus.fuoye.ng
   B. www.fuoye.org.ng
   C. www.fuoye.edu.com
   D. www.fuoye.edu.ng

   ✔ **Answer: D.** The main site of the university is fuoye.edu.ng.

2. **What do we call the skills that guide library users to make full use of library resources like books and journals?**

   A. Study skills
   B. Reading skills
   C. Library skills
   D. Listening skills

   ✔ **Answer: C.** Library skills help users get the most out of the library.

3. **What is the full meaning of SQ3R?**

   A. Survey, Question, Read, Recite and Review
   B. Survey, Query, Read, Recite and Review
   C. Solution, Question, Recite, Read and Review
   D. Solve Query, Recite, Read, Review

   ✔ **Answer: A.** SQ3R = Survey, Question, Read, Recite, Review.

4. Which one is an educational resource centre, a store house of knowledge and a nest where scholars are hatched?

A. School
B. Store house
C. Academic
D. Library

✔ **Answer: D.** The library is the store house of knowledge and a nest where scholars are hatched.

5. All of these are reasons for studying use of library except one. Which one?

A. It is needful and helpful to become a successful scholar
B. It acquaints you with library resources
C. It is needed to meet peers
D. It is a requisite course for all undergraduates

✔ **Answer: C.** Use of Library is not about meeting peers; the other three are given in the course.

6. Libraries attached to tertiary institutions like a university, polytechnic or college of education are called what?

A. Public libraries
B. National libraries
C. Academic libraries
D. Private libraries

✔ **Answer: C.** FUOYE Library and EKSU Library are academic libraries.

7. What is the apex library of any nation called?

A. Private library
B. Public library
C. National library
D. Special library

✔ **Answer: C.** The national library has the legal deposit right.

8. What is the full meaning of ISSN?

A. International Standard Serials Number
B. Internal Standard Serial Number
C. Interrelation Standard Serial Number
D. International Serials Standard Numbers

✔ **Answer: A.** ISSN identifies serials.

9. What is the full meaning of ISBN?

A. International Serial Board Number
B. Inter-relation Serial Book Number
C. International Standard Board Number
D. International Standard Book Number

✔ **Answer: D.** ISBN identifies one book title in the whole world.

10. FUOYE Library is an example of which type of library?

A. Academic library
B. School library
C. Public library
D. Private library

✔ **Answer: A.** It is attached to a university.

11. Which term describes the technology that uses computers, communication devices and software to store, process, transmit and retrieve information anywhere, anytime?

A. Information and Communication Technology (ICT)
B. Information Technology (IT)
C. Information Processing Technology
D. Internet and Communication Tools

✔ **Answer: A.** ICT is wider than IT because it also covers communication.

12. Which site is used for students' registration, courses and payment activities (the university portal)?

A. www.fuoye.edu.ng
B. www.fuoye.edu.ng/payments
C. www.ecampus.fuoye.edu.ng/portal
D. www.fuoye.edu.ng/portal

✔ **Answer: C.** Students register, do courses and pay fees on the portal.

13. What is the full meaning of SSID?

A. Service Set Identifier
B. Servicecom Set Identification
C. Servicecom Set Identifiers
D. Service Set Identification

✔ **Answer: A.** An SSID is the name of a Wi-Fi network.

14. What is the act of transmitting messages called?

A. Communication Technology
B. Transmitter
C. Communication
D. Information

✔ **Answer: C.** Communication = sending and receiving messages.

15. What do we call the use of scientific knowledge, experience and resources to create processes and products that meet human needs?

A. Communication
B. Information Technology based
C. Technology
D. Information

✔ **Answer: C.** That is the definition of technology.

16. Someone steals your account information from a trusted bank. This is an example of what?

A. Computer practice
B. Computer threats
C. Computer act
D. Computer ethics

✔ **Answer: B.** Stealing account information is a human threat to computers.

17. Which program protects a computer against viruses by finding and removing them from memory, storage media and incoming e-mail files?

A. Threat removal
B. Virus removal
C. Computer virus
D. Antivirus

✔ **Answer: D.** Examples: Norton, ESET and Kaspersky.

18. Which piece of hardware or software stops communications forbidden by the security policy in a network?

A. Malware
B. Adware
C. Spyware
D. Firewall

✔ **Answer: D.** A firewall enforces the security policy.

19. What is the full meaning of WAN?

A. Wireless Area Network
B. Wireless Access Network
C. Wide Area Network
D. Wider Access Network

✔ **Answer: C.** A WAN spans long distances.

20. What is the full meaning of LAN?

A. Local Area Network
B. Land Area Network
C. Land Access Network
D. Local Access Network

✔ **Answer: A.** A LAN covers one building or campus.

21. Taking another person's idea and copying it as if it were your own is called what?

A. Counterfeiting
B. Photocopy
C. Plagiarism
D. Piracy

✔ **Answer: C.** Plagiarism is a grievous offence in academia.

22. What is the exclusive right given to an author or creative artist to authorise the use of their work in any way?

A. Patent
B. Trademark
C. Royalty
D. Copyright

✔ **Answer: D.** Copyright is shown with the (c) symbol.

23. Which libraries are owned by an individual or an organisation like a church, built around the owner's field?

A. Private libraries
B. Public libraries
C. Special libraries
D. School libraries

✔ **Answer: A.** Their collections are dictated by the owner's discipline.

24. Who defined serials as publications issued in successive parts, at regular or irregular intervals, meant to continue indefinitely?

A. Dr. Mrs Olubanke Bankole
B. Osborn (1980)
C. Dr. Akinwunmi
D. Osborn (1970)

✔ **Answer: B.** This is the definition quoted in the lecture notes.

25. Publications issued in successive parts and meant to continue indefinitely, like newspapers and journals, are called what?

A. Cataloguing
B. Serials
C. Reference
D. Periodical

✔ **Answer: B.** Serials include journals, newspapers, conference proceedings and bulletins.

26. A book with fewer than 49 pages is known as what?

A. Journal
B. Index
C. Pamphlet
D. Serial

✔ **Answer: C.** 49 pages or more is a book.

27. What is the address for FUOYE email?

A. support@fuoye.edu.ng
B. gmail@fuoye.edu.ng
C. www.fuoye.edu.ng
D. support.fuoye.edu.ng

✔ **Answer: A.** This is the support mail given in the past questions.

28. All of these are departments in the library except one. Which one?

A. Teaching Sector
B. Readers' Services department
C. Technical Services department
D. Administrative department

✔ **Answer: A.** Administrative, Technical Services and Readers' Services are library departments.

29. What is the process of describing a book or library item so that its important bibliographic details are highlighted?

A. Serials
B. Cataloguing
C. Bibliography
D. Description

✔ **Answer: B.** The person who does it is the cataloguer.

30. All of these are types of catalogue except one. Which one?

A. Book catalogue
B. Shelf catalogue
C. Reserve catalogue
D. Card catalogue

✔ **Answer: C.** Types: card, OPAC, book, shelf, CD-ROM and COM catalogues.

31. What does LCCS stand for?

A. Library of Classification Congress Schedule
B. Library of Congress Classification Scheme
C. Library and Communication Congress Schedule
D. Library of Classification Congress Scheme

✔ **Answer: B.** It is the scheme used by academic libraries.

32. All of these are importance of serials except one. Which one?

A. They are used to highlight text
B. They carry first information on new discoveries
C. They contain up-to-date information
D. Their information may not appear in books

✔ **Answer: A.** Serials are primary sources of up-to-date information, not tools for highlighting.

33. Where is the registration of new library users done?

A. Circulation section
B. Library attendance desk
C. ICT section
D. Personnel office

✔ **Answer: A.** Circulation also handles loans, overdue notices and fines.

34. In which section of the library are reference sources kept?

A. Circulation section
B. Reserve section
C. Serials section
D. Reference section

✔ **Answer: D.** Reference sources are consulted there and not loaned out.

35. Reference sources are easily identified by which marking on the spine?

A. "Ref" or "R"
B. RF
C. REFs
D. REFER

✔ **Answer: A.** Reference books carry R or Ref on the spine.

36. All of these are examples of reference sources except one. Which one?

A. Encyclopedias
B. Dictionaries
C. Identification of files
D. Biographies and gazettes

✔ **Answer: C.** Indexes, abstracts, encyclopedias, dictionaries, biographies and gazettes are reference sources.

37. What is the list of all items in a collection, arranged by author, title or subject, called?

A. Register
B. Catalogue
C. Finding aid
D. Bibliography

✔ **Answer: B.** It is an index to the collection of a library.

38. All of these are library rules and regulations except one. Which one?

A. Registered students get library cards to bring along
B. Only registered students may use the library
C. All registered students may take books out of the library
D. Students must dress decently according to the FUOYE dress code
✔ **Answer: C.** Reference, serials and reserved materials cannot be taken out of the library.

39. What is the full meaning of OPAC?

A. Online Public Area Catalogue
B. Online Private Access Cataloguing
C. Online Public Access Catalogue
D. Online Publication Access Catalogue
✔ **Answer: C.** OPAC is the most modern form of catalogue.

40. What do we call an online database of materials held by a library or group of libraries?

A. OPAC
B. Card catalogue
C. Shelf catalogue
D. Book catalogue
✔ **Answer: A.** OPAC = Online Public Access Catalogue.

41. Which software packages are designed to perform and accomplish library functions?

A. University
B. Library
C. Catalogue
D. Shelf

✔ **Answer: B.** Library software handles cataloguing, circulation, acquisition and serials.

42. Which term means grouping library materials by subject in a logical order?

A. Indexing
B. Accessioning
C. Cataloguing
D. Classification

✔ **Answer: D.** Classification is a way of grouping things.

43. To locate a book on the shelf, you need what?

A. Call mark from Koha
B. Serials list
C. Library card
D. Catalogue page

✔ **Answer: A.** The call mark directs the user to where the book sits on the shelf.

44. Which term is otherwise known as audio-visual materials?

A. Booklet materials
B. Non-book materials
C. Graphics
D. Templates

✔ **Answer: B.** Films, slides, TVs, radios and computers are non-book materials.

45. All of these are examples of non-book materials except one. Which one?

A. Television
B. Paper
C. Computer
D. Slides

✔ **Answer: B.** Paper is not an audio-visual item.

46. Which of these means the same as library users?

A. Patrons
B. Vendors
C. Donors
D. Personnel

✔ **Answer: A.** Clientele, users, readers and patrons are used interchangeably.

47. Where do users borrow books from the library?

A. Reference section
B. Serials section
C. Reserve section
D. Circulation section

✔ **Answer: D.** Materials are charged and discharged at the circulation section.

48. Which department is called the backbone of a library?

A. Readers' Services department
B. Technical Services department
C. Electronic department
D. ICT department

✔ **Answer: B.** Technical Services does the 'behind the scene' work.

49. Which two things are required to access library internet resources?

A. Username and password
B. Matric number and username
C. Face and thumbprint
D. ID number and password

✔ **Answer: A.** A login is needed for e-resources.

50. All of these are types of operating system except one. Which one?

A. Unix
B. Windows
C. Linux
D. Infinix

✔ **Answer: D.** Infinix is a phone brand, not an operating system.

51. To whom should the loss of a library card be reported?

A. The chief cataloguer
B. The ICT manager
C. The serials librarian
D. The circulation librarian

✔ **Answer: D.** Circulation handles users' cards and registration.

52. Theft of library materials can lead to what?

A. A pardon
B. Suspension or expulsion
C. A reward from the library
D. A free replacement of the book

✔ **Answer: B.** Sanctions include suspension and dismissal after trial.

53. All of these are steps in how books get to the shelf except one. Which one?

A. Accessioning
B. Decoding
C. Stamping
D. Cataloguing and classifying

✓ **Answer: B.** Steps: stamp, accession, catalogue and classify, assign call mark, shelve.

54. A classification number joined to a cutter number, like 131 and B49, is called what?

A. Barcode
B. Accession number
C. ISBN
D. Call mark

✓ **Answer: D.** The call mark is pasted on the spine of the book.

55. Which piece of code is put into an operating system or software and does harm after a set time or when conditions are met?

A. Firewall
B. Logic bomb
C. Antivirus
D. Computer bomb

✓ **Answer: B.** A time bomb is an example of a logic bomb.

56. Logic bombs are often used with viruses, worms and trojan horses to do what?

A. Block them using the firewall policy
B. Remove them from the computer before they spread
C. Time them to do damage before being noticed
D. Back up files before the damage starts

✔ **Answer: C.** They are timed for maximum damage before detection.

57. Where is the new FUOYE library located?

A. Phase one
B. Phase three
C. Phase two
D. Administrative block

✔ **Answer: B.** This is the answer given in the past questions.

58. The approach used when searching for Catalog Material is called ___?

A. None of the above
B. ISBN
C. KOHA
D. Cataloguing

✔ **Answer: C.** This is the answer given in the past questions.

59. Quota numbers direct a user to ___?

A. The Information needed
B. The Catalogue page
C. Note
D. The Book shelve

✔ **Answer: A.** This is the answer given in the past questions.

60. Worm in relation to ICT refers to ___?

A. Adware
B. Virus
C. None of the above
D. Antivirus

✔ **Answer: B.** This is the answer given in the past questions.

LAST-MINUTE SHEET

One-Page Hack Sheet

Read this on the way to the hall.

Chapter 1: Use of Library (Basics, Types, Stock & Rules)

- Types of library = APN-PRS: Academic, National, Private, Public, Research/Special, School. Six, never seven.
- Four components = B-B-P-C: Books/materials, Building, Personnel, Clientele. If an option names only 3 of them, it is incomplete.
- Four ways to acquire stock = P-G-E-B: Purchase, Gift, Exchange, Bequeath. Gift comes free, Purchase is cash and carry.

Chapter 2: Serials, Reference Sources & Intellectual Property Rights

- Directional = DIRECTS you elsewhere = Bibliographies, Indexes, Abstracts (B-I-A). Everything else (dictionary, atlas, almanac...) gives the answer directly.
- Thesaurus gives words that mean the same or opposite but NEVER meanings. Dictionary gives meanings. Don't swap them.
- Plagiarism = copying ideas without credit. Piracy = copying for money. Counterfeit = fake copy of the original. Money in the picture means piracy.

Chapter 3: Book Anatomy, Citation & Referencing

- Book has TWO major parts: Preliminary pages (before the text, roman numerals) and the Body/Text. Preface,

dedication, acknowledgement, table of contents all sit in the preliminary part.

- Verso = back of the title page (copyright details). Recto = right-hand page (odd number). V comes before R, back before right.
- Harvard = author-date. MLA = author + PAGE number, no year in the bracket. APA = author + year. Page number in the bracket points to MLA.

Chapter 4: Catalogue, Classification & Sections of the Library

- LC for Academic libraries, Dewey for National and Public libraries. Think: L-C = Learned Campus, Dewey = the people's library.
- Dewey: 10 classes, each 100 numbers wide. 500s = pure science, 600s = technology. Science sub-order is M-A-P-C-G-P-A-B-Z: Maths 510, Astronomy 520, Physics 530, Chemistry 540, Geology 550... Zoology 590.
- Technical Services = 'behind the scene' (Acquisition, Cataloguing, Serials, Reprography/Bindery). Readers' Services = where users meet staff (Circulation, Reserve, Reference).

Chapter 5: Library Software & Electronic Databases

- Match the database to the first letter of its field: HINARI = Health (WHO), AGORA = Agriculture (FAO), OARE = Environment (UNEP), TEEAL = Agriculture (offline), DOAJ = Open access journals, PubMed = Biomedical.
- Two software types: System (runs the machine: Windows, Linux, Android) and Application (does a job for the user: Word, Photoshop, library software). Library software is application software.

- Proprietary = you pay (Symphony, Millennium, Aleph, Voyager). Open source = free to use, modify and share (Koha, Evergreen, Greenstone, DSpace). FUOYE runs Koha.

Chapter 6: Sources of Information (Primary, Secondary & Tertiary)

- PRIMARY = first-hand, raw, original (diary, letter, speech, patent, thesis, interview, artefact). SECONDARY = someone talking ABOUT the original (textbook, commentary, criticism). TERTIARY = points you to the others (bibliography, index, directory, chronology).
- Quick test: did the writer see it happen? Yes = primary. Did the writer read about it? = secondary. Does it just list where to look? = tertiary.
- Textbook, dictionary and encyclopedia are secondary. Bibliography, index and directory are tertiary. Don't swap the two lists.

Chapter 7: Online Searching (Challenges, Strategies & Evaluation)

- Seven challenges = S-I-E-A-N-H-R: Spelling variation, Information overload, Excessive truncation, Acronyms, Negation, Homographs, Reverse concept. Tie each to its example: honour/honor, too general, LIB, ADD/SAD/AIDS, elderly, cricket/KM, school management vs management school.
- Boolean: AND narrows (both words), OR widens (either word), NOT removes a word. Truncation too short (LIB) pulls unrelated words.
- Evaluating a webpage = T-A-P-C-D: Title, Author (name and designation), Purpose, Contact details, Date (produced and last updated).

Chapter 8: Deterioration, Preservation & Conservation

- Three agents of deterioration = E-B-M: Environmental, Biological, Microbiological. Biological = visible (humans, insects). Microbiological = needs a microscope (fungi, bacteria, mould).
- Preservation = protect everything that keeps the material healthy (broad, preventive). Conservation = specialised repair to make a fragile item safe and usable (hands-on).
- Five inhibitors = F-M-G-M-T: Finance, Materials, Government economic policies, Maintenance culture, Technical know-how. Money problem comes first.

Chapter 9: ICT: Threats, Networks, Internet, IoT & NFTs

- Malicious code family: Virus (attaches to files, needs a host), Worm (stands alone, travels through the network), Trojan horse (looks useful, tricks you into installing), Logic bomb (waits for a condition or date), Trapdoor/Backdoor (secret entry). Worm = Network, Virus = Files.
- Three network sizes to remember: LAN (one building or campus), CAN (whole campus, e.g. fuoyeNET), WAN (long distance, the Internet is a super WAN). Topologies are Bus, Ring, Star.
- NFT = unique, cannot be altered, one-of-a-kind (Non-Fungible). Cryptocurrency = digital money. Blockchain = the shared ledger that records both. Buying an NFT does not mean you own the copyright.

Chapter 10: Computers: Classification, Components & Supercomputers

- Timeline in order: Abacus (500 BC) > Napier's Bones (1617) > Slide Rule (1632) > Pascaline (1642) > Leibniz > Jacquard loom > Babbage Difference Engine (1822) > Analytical Engine (1833) > Hollerith (1890) > ENIAC.
- Four ways to classify a computer = G-T-S-P: Generation, Types (analog, digital, hybrid), Size (super, mainframe, mini, micro), Purpose (special, general).
- Top 5 supercomputers in order of speed: Fugaku (Japan) > Summit (USA) > Sierra (USA) > Sunway TaihuLight (China) > Perlmutter. Japan, USA, USA, China, USA.

FUOYE: FUOYE Website, Portal, Email & Campus Network

- Portal sites (registration, courses, payments) = ecampus.fuoye.edu.ng and portal.fuoye.edu.ng. Main website = fuoye.edu.ng. Email ends @fuoye.edu.ng. Wi-Fi name = fuoyenet.
- Default email password is fuoye123, and you MUST change it. Login format has your name and matric number inside, in a set order.
- FUOYE library runs Koha (open source). The campus network (CAN) only residents can use is called fuoyeNET.